

No plan, No Aid? The effects of National Adaptation Plan adoption on received Adaptation Aid*

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Abstract

This paper examines how National Adaptation Plan (NAP) adoption affects the adaptation finance a country receives. We situate donor allocation within the IPCC AR6 risk framework. The adaptation-finance literature has operationalized recipient need through physical hazard and through composite vulnerability indices that fuse hazard, exposure, and adaptive capacity, leaving the institutional adaptive-capacity component unidentified. We model NAP adoption as an information shock that updates donors' perception of a recipient's adaptive capacity. Holding need fixed (hazard via country fixed effects; population and baseline capacity as controls), we provide the first causal identification of the allocative role of this component. Exploiting the staggered timing of NAP adoption in a difference-in-differences design, we find that NAP adoption raises adaptation commitments by about 35% and a country's share of global adaptation finance by 0.3 percentage points, concentrated among DAC bilateral and multilateral donors.

Keywords: Adaptation aid; climate finance; National Adaptation Plans; development economics; staggered difference-in-differences.

JEL codes: F35; O19; Q54; Q58.

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1 Introduction

A central question in development economics is how donors allocate scarce aid across recipients that differ in both need and institutional capacity. The aid-allocation literature usually distinguishes between need-based motives, which direct resources toward poorer or more vulnerable countries, and merit-based motives, which favor recipients with stronger policies, institutions, or implementation capacity ([Alesina and Dollar, 2000](#); [Burnside and Dollar, 2000](#); [Hoeffler and Outram, 2011](#); [McGillivray, 2004](#)). This distinction matters because it governs how observed flows should be interpreted, whether a given allocation reflects need, merit, or both. In practice, however, need and capacity are often negatively related. Donors, therefore, face a trade-off between allocating resources to where needs are greatest and allocating them to where projects are expected to be implemented most effectively. This paper studies whether a domestic policy reform can shift this trade-off by changing donors' beliefs about recipient capacity. This shifts the question from explaining which countries receive more aid to asking whether recipient countries can change donor beliefs, and hence donor allocations, through credible planning reform.

Climate adaptation finance is a natural context in which to study this mechanism. Adaptation aid changes the meaning of recipient need: rather than being defined primarily by income or poverty, need is increasingly framed in terms of climate vulnerability. This vulnerability is multidimensional, combining physical exposure to climate hazards, socio-economic sensitivity, and the capacity to cope with and adapt to climate impacts. Recent work has shown that these dimensions need not play the same allocative role. In particular, [Weiler et al. \(2018\)](#) find that donors partly respond to physical exposure, while low adaptive capacity does not attract more adaptation aid; instead, donors also reward good governance and pursue economic interests. These findings suggest that adaptation finance reproduces the classic need–merit dilemma in a climate-specific form: donors may wish to prioritize countries facing greater climate risks, but they may also prefer recipients with stronger planning and implementation capacity. We build on this insight by shifting the empirical object from cross-country differences in exposure, adaptive capacity, or governance to a dated policy signal. We ask whether the same country receives more adaptation finance after adopting a National Adaptation Plan, a reform that plausibly updates donors' perceptions of adaptation planning capacity while leaving physical exposure unchanged.

National Adaptation Plans (NAPs) provide a clear setting to test whether such policy signals shift adaptation finance. Introduced under the UNFCCC ([UNFCCC, 2011](#)), NAPs are medium-term planning instruments that formalize adaptation strategies and coordinate sectoral policy. They are explicitly framed as tools to unlock adaptation finance, and donors such as the World Bank report using them to organize adaptation-related financing ([UNFCCC, 2023](#); [World Bank, 2019](#)). NAP adoption is therefore a discrete, dated event that plausibly updates donors' perceptions of a recipient's adaptive capacity while leaving its physical hazard unchanged. It

constitutes an information shock to the merit margin, holding fixed time-invariant physical exposure and controlling for slow-moving need proxies. We therefore estimate whether NAP adoption causally affects the volume and composition of adaptation-related development finance received by developing countries.

The expected effect is theoretically ambiguous. If donors reward capacity, NAP adoption should raise adaptation finance by increasing perceived aid efficiency; if they prioritize adaptation need, a structured plan could instead lower perceived urgency and redirect resources toward non-adopters. The net effect depends on how donors trade off need against capacity, both within countries, through the adaptation share of a given aid envelope, and across countries, through the size of the envelope allocated to each recipient. We clarify these channels with a simple recipient–donor allocation model in which a donor allocates a fixed global budget across recipients and, within each recipient, between adaptation and other aid. NAP adoption enters as a policy-driven increase in effective capacity and generates two empirically separable channels: a composition channel, in which higher capacity raises the adaptation share within a country’s envelope, and an envelope channel, in which higher perceived returns expand the total envelope allocated to adopters. The model yields testable predictions for adaptation finance, total finance, non-adaptation finance, global adaptation shares, and donor-type heterogeneity.

Empirically, we assemble a novel country–year panel linking the timing of NAP adoption to project-level adaptation finance from the OECD Creditor Reporting System (CRS), identified via the climate-adaptation Rio markers, over 2009–2024. Exploiting staggered adoption, we estimate dynamic treatment effects using the [Callaway and Sant’Anna \(2021\)](#) estimator for staggered difference-in-differences designs, and corroborate the results with [de Chaisemartin and d’Haultfoeuille \(2023\)](#). We compare adopters to never-treated countries, control for pre-treatment institutional quality (WGI Government Effectiveness) and (log) population evaluated at $g - 1$, report event-time leads and lags, and conduct placebo tests and sensitivity analyses, including HonestDiD bounds following [Rambachan and Roth \(2023\)](#).

Our findings provide evidence that NAP adoption affects the volume and allocation of adaptation-related commitments through a dual mechanism. Adaptation commitments rise ($\approx 34\%$, $p < 0.05$) while total commitments are statistically flat and non-adaptation flows remain flat (the pre-trend evidence for these two secondary outcomes is weaker, a caveat we document in Section 5.2). This pattern points to a composition channel — adaptation’s share within the aid envelope rises — rather than an expansion of the total envelope. We also find that adopters’ shares of global adaptation finance rise by approximately 0.3 percentage points after adoption ($p < 0.05$), robust to cohort inclusion. Heterogeneity analyses reveal that the adaptation commitment effect is driven by DAC bilateral ($\approx 36\%$, $p < 0.05$) and multilateral donors ($\approx 74\%$, $p < 0.05$), while non-traditional donors show no response, consistent with NAPs functioning as readiness signals for capacity-oriented public donors. The effect is concentrated in countries with below-median governance capacity ($p < 0.05$), though a pre-trend concern in that subgroup warrants caution. LDC and non-LDC subsamples both yield imprecise estimates,

most plausibly because the LDC estimation sample contains only 17 countries. Among income groups, low-income and lower-middle-income adopters show positive effects ($\approx 38\text{--}52\%$).

Our paper makes three contributions. First, we contribute to the aid-allocation literature by treating recipient merit not only as an observed country characteristic, but as a policy-shifted signal. Existing studies commonly proxy merit using governance, policy, or institutional indicators; we instead study a dated domestic reform that plausibly changes donor perceptions of implementation capacity. This allows us to ask whether recipient governments can move along the need–capacity frontier by producing credible planning institutions. Second, we contribute to the literature on adaptation-finance allocation by turning its decomposition of climate vulnerability into an identification strategy. Existing studies show that physical exposure, adaptive capacity, governance, and donor interests play different roles in adaptation-aid allocation. We build on this insight by using the IPCC risk framework to map physical risk components to recipient need and adaptive capacity to recipient merit, and by exploiting NAP adoption as a dated policy signal that shifts the latter while leaving the former unchanged. Third, we provide quasi-experimental evidence on the allocative role of donor-perceived adaptive capacity. Using a new country–year panel that links NAP adoption timing to project-level adaptation finance, and exploiting staggered NAP adoption with modern difference-in-differences methods, we estimate how a discrete signal of adaptation readiness affects the volume, composition, and global distribution of adaptation finance.

The remainder of the paper proceeds as follows. Section 2 reviews the aid-allocation literature and develops our risk decomposition. Section 3 presents the allocation model and derives testable hypotheses. Section ?? describes the data and empirical strategy. Section ?? reports results, dynamics, robustness, and HonestDiD sensitivity. Section ?? concludes.

2 Related literature

2.1 Aid allocation and the need–merit dilemma

The allocation of development aid has long been understood through three motives: recipient need, donor interest, and recipient merit. Early development assistance was normatively grounded in altruism, with donors aiming to support the countries most in need, typically defined by low income (Lancaster, 2008). A large empirical literature subsequently challenged this altruistic view. Aid flows are not allocated solely according to recipient need; they also respond to donor self-interest, including strategic alliances, commercial relationships, geopolitical objectives, and postcolonial ties (Alesina and Dollar, 2000; Hoeffler and Outram, 2011; Faye and Niehaus, 2012). This donor-interest motive matters because observed aid flows may reflect donors’ own objectives as much as recipients’ needs.

Since the 1990s, reflecting post-Washington-consensus thinking (Kanbur, 2000), a recipient-merit model has emerged. Here, donors prioritize countries with good governance and effective

institutions on the premise that aid is more effective in capable hands ([Burnside and Dollar, 2000](#)). This trend tracks the rise of aid-effectiveness norms embodied in the Paris Declaration (2005) and the Accra Agenda for Action (2008). The merit rationale, however, introduces its own dilemma: institutional quality is rewarded as a safeguard against inefficiency, but this can bypass the countries with the weakest institutions, often those most in need, in favor of more “deserving” partners.

Donor heterogeneity is therefore central to our argument. A signal of recipient capacity should matter only if donors use capacity when allocating aid. Donors with formal appraisal procedures are more likely to value credible planning and implementation capacity, while more demand-driven donors may respond less to such signals and leave greater room for recipient-side discretion. Evidence from the aid-allocation literature supports this logic. In a direct comparison of donor procedures, [Dreher et al. \(2019\)](#) show that World Bank aid is not disproportionately directed toward leaders’ birth regions, whereas recipient-initiated Chinese aid is. The same pattern appears at the cross-country level: Chinese aid is less conditioned on recipient governance than aid from some traditional donors ([Dreher and Fuchs, 2015](#); [Dreher et al., 2018](#)). Together, these findings suggest that donors differ not only in how much aid they provide, but also in how strongly they condition aid on recipient capacity. In our context, this implies that NAP adoption should affect donors differently, with stronger responses among donors for whom recipient planning capacity is part of the allocation rule.

These three traditions, i.e., recipient need, donor interest, and recipient merit, share an analytical strength. Because the motives are conceptually distinct, an allocation regression can, in principle, ask whether observed flows respond to recipient need, donor objectives, or recipient capacity ([Hoeffler and Outram, 2011](#)). Adaptation finance retains this framework but makes the need–merit boundary harder to draw. Climate vulnerability includes physical exposure, which maps naturally onto need, but also adaptive capacity, which is closely related to the merit and absorptive-capacity concerns of the aid-allocation literature. The next subsection shows how this ambiguity has shaped the adaptation-finance literature and why a dated policy signal such as NAP adoption can help isolate the capacity margin.

2.2 The need–merit dilemma in climate adaptation finance

Climate finance has inherited, and is now reshaping, classic models of aid distribution ([Betzold and Weiler, 2018](#); [Weiler et al., 2018](#)). Adaptation aid changes the meaning of recipient need: rather than being defined primarily by income or poverty, need is increasingly framed in terms of climate vulnerability. Under the UNFCCC, developed countries agreed to assist developing countries that are “particularly vulnerable” to the adverse effects of climate change, making vulnerability a central criterion for claims over scarce adaptation resources ([Weiler et al., 2018](#)). This shift is not merely semantic. It changes the object of allocation from poverty alone to a multidimensional concept that includes physical exposure to climate hazards, socio-economic

sensitivity, and the capacity to cope with and adapt to climate impacts (Füssel, 2007; Hinkel, 2011).

The adaptation-finance literature has already recognized that these dimensions need not play the same allocative role. In a leading contribution, Weiler et al. (2018) explicitly bring the aid-allocation framework to adaptation finance by asking whether bilateral adaptation aid responds to recipient need, recipient merit, or donor interests. Their results show that donors partly respond to physical vulnerability: countries more exposed to climate risks are more likely to receive adaptation aid and receive more adaptation aid per capita. Yet low adaptive capacity does not attract more aid. Instead, donors also reward good governance and pursue economic interests, while adaptation aid closely follows existing development-aid relationships. These findings are important because they show that climate vulnerability is not a single allocation motive. Its physical components resemble need, whereas adaptive capacity is closely related to the merit and absorptive-capacity concerns familiar from the development-aid literature.

Our paper builds directly on this insight but shifts the empirical object. Existing allocation studies ask whether countries with different levels of exposure, adaptive capacity, governance, or donor ties receive different amounts of adaptation aid. We ask whether the same country receives more adaptation finance after adopting a National Adaptation Plan, a dated domestic policy reform that plausibly updates donors' perceptions of adaptation planning and implementation capacity while leaving physical exposure unchanged. This allows us to study donor-perceived adaptive capacity as a policy-shifted margin, rather than as a slow-moving country characteristic embedded in vulnerability or governance indicators.

This distinction matters for both identification and theory. If donors treat planning capacity as a merit signal, NAP adoption should increase adaptation finance, especially among donors whose allocation rules make project preparation, readiness, and implementation capacity salient. If donors instead prioritize physical need, NAP adoption should have little effect, or could even redirect resources toward non-adopters perceived as more vulnerable. We therefore use the IPCC risk framework to keep separate the physical components of climate risk from the institutional capacity margin shifted by NAP adoption, and we formalize the resulting allocation channels in the next section.

2.3 From vulnerability components to an identification framework

The preceding evidence shows why adaptation finance raises a sharper version of the standard need–merit dilemma. Physical exposure to climate hazards maps naturally onto recipient need, while adaptive capacity is simultaneously part of climate vulnerability and closely related to the merit and absorptive-capacity concerns of the development-aid literature. To study NAP adoption, we therefore need a framework that separates these components and clarifies which one can plausibly be shifted by a dated policy reform.

The IPCC risk framework provides this structure. In its Third and Fourth Assessment

Reports (TAR/AR4) (IPCC, 2001, 2007), the IPCC conceived vulnerability as the joint product of exposure, sensitivity, and adaptive capacity (Füssel, 2007). This tri-dimensional construct has proved hard to operationalize econometrically, because adaptive capacity is itself shaped by institutions, income, and governance, so that composite indices can mix physical, socio-economic, and institutional dimensions (Hinkel, 2011). The IPCC subsequently moved toward a more disaggregated risk framework. In SREX (IPCC, 2012) and in AR5 and AR6 (IPCC, 2014, 2022), exposure is treated as an analytically distinct component, and risk is defined as a function of hazard, exposure, and vulnerability, with vulnerability narrowed to sensitivity and the capacity to cope and adapt (Oppenheimer et al., 2014). This shift is useful for our purposes because it separates physical exposure from the institutional and socio-economic dimensions through which countries manage climate risk.

A similar separation underpins much of the economics of climate and development. As synthesized by Dell et al. (2014), this literature identifies climate–economy relationships from exogenous physical variation in temperature, precipitation, cyclone intensity, or sea level (Kahn, 2005; Barrios et al., 2010; Felbermayr and Gröschl, 2014; Burke et al., 2015; Hallegatte et al., 2020). Two methodological moves are especially relevant. Kahn (2005) decomposes disaster mortality into physical hazard, income, and institutions, treating physical exposure as analytically separable from socio-economic and institutional capacity. Felbermayr and Gröschl (2014) similarly construct physical hazard indicators as an alternative to loss-based measures, arguing that loss-based indicators confound the geophysical intensity of events with countries’ socio-economic capacity to absorb them. Applied to adaptation aid, Weiler et al. (2018) follows the same logic empirically by showing that donors respond more strongly to physical exposure than to composite vulnerability or low adaptive capacity.

These two movements, i.e., the IPCC’s top-down separation of risk components and the climate-economics literature’s bottom-up isolation of physical hazards—converge on the same partition: a physical layer, comprising hazard and exposure, and a socio-economic and institutional layer, comprising sensitivity and adaptive capacity. We use this partition to map each risk component onto the allocation motive emphasized in the development-aid literature, as summarized in Table 1. The organizing principle is temporality. Time-invariant physical components enter the model as primitives absorbed by country fixed effects; slow-moving demographic and socio-economic components enter as controls or are absorbed by fixed effects; and the component plausibly shifted by policy, i.e., donor-perceived adaptive capacity, is the margin on which NAP adoption operates. This decomposition is what links the adaptation-finance literature to our empirical design: it allows us to interpret NAP adoption not as a shock to climate need, but as a dated signal of implementation capacity.

Table 1: Mapping the IPCC AR6 risk object onto development-aid allocation roles

IPCC AR6 component	Development-aid role	Time profile	Model object	Empirical handling
Hazard	Need (geophysical)	Time-invariant	H_i (primitive)	Country FE
Human exposure	Need (demographic)	Slow-moving	X_{it} (population)	Control
Socio-economic sensitivity	Need (socio-econ.)	Time-varying	X_{it}	Control / FE-absorbed
Adaptive capacity	Merit (institutional)	Policy-shifted	V_i (primitive, NAP)	Identified (β_e)

Notes: The IPCC Sixth Assessment Report (IPCC, 2022) defines climate risk as a function of hazard, exposure, and vulnerability, with vulnerability restricted to sensitivity and the (lack of) capacity to cope and adapt. Each row assigns one risk component to the need or merit motive that the development-aid allocation literature treats as analytically distinct. “Time profile” is the organizing principle of the assignment: time-invariant need components enter the model as primitives absorbed by country fixed effects (FE); time-varying need components enter as controls X_{it} (here proxied by log population); the policy-shifted merit primitive V_i (adaptive capacity), upgraded by NAP adoption, is the channel identified by the event-study coefficients β_e . Donor interest, the third allocation motive, is not a risk component and is addressed through fixed effects and donor-type heterogeneity (Section 5.8). V_i denotes donor-perceived adaptive capacity, the favorable pole of the vulnerability dimension (high V_i corresponds to low vulnerability).

3 A Recipient–Donor Allocation Model with NAP

We develop a simple theoretical framework not to predict aid levels per se, but to clarify how NAP adoption may affect adaptation finance through distinct and empirically separable channels.

3.1 Primitives and states

Each recipient country i is characterized by three objects: a structural hazard environment $H_i \geq 0$, a baseline adaptive capacity $x_i \geq 0$, and a National Adaptation Plan indicator $A_i \in \{0, 1\}$. These objects operationalize the conceptual mapping of Section 2.3. The hazard term H_i captures the geophysical component of recipient need, while adaptive capacity captures the institutional component through which donors assess whether adaptation projects can be designed, coordinated, and implemented effectively.

Hazard (H_i). We define H_i as country i ’s structural hazard environment: a persistent geophysical primitive shaped by geography and physical conditions, such as coastline length, elevation, aridity, cyclone exposure, and dependence on rain-fed agriculture. Consistent with the decomposition in Section 2.3, H_i captures the physical hazard component of need and is distinct from human exposure, socio-economic sensitivity, and adaptive capacity. The remaining AR6 risk components are decomposed across the model’s analytical roles rather than folded into H_i :

- Human exposure and socio-economic sensitivity are the demographic and socio-economic need components, including population, income, poverty, and sectoral composition. In the empirical specification, these components are captured by observables X_{it} , with their

time-invariant part absorbed by country fixed effects. They are need components held fixed by the design, not excised from need.

- Adaptive capacity is the institutional and organizational dimension on which NAP adoption operates. Keeping it distinct from the need components H_i and X_{it} allows the design to identify the capacity channel: folding it into H_i would make NAP adoption shift need and readiness at the same time.

Adaptive capacity (x_i , V_i) and the NAP signal. Baseline adaptive capacity x_i captures the recipient’s pre-treatment ability to design, coordinate, and implement adaptation interventions; empirically we proxy it by pre-treatment institutional quality, the World Governance Indicators’ Government Effectiveness score at $g - 1$. National Adaptation Plan adoption upgrades donor-perceived capacity multiplicatively:

$$V_i = (1 + \delta_i A_i) x_i, \quad \delta_i > 0. \quad (1)$$

We interpret δ_i as the information content of the NAP signal perceived by donors: adopting a formal planning instrument conveys coordination capacity and a credible commitment to adaptation, reducing donors’ uncertainty about a recipient’s ability to use funds effectively, consistent with donors’ stated use of NAPs as eligibility and programming anchors ([World Bank, 2019](#)). We allow δ_i to be larger when baseline capacity x_i is lower, reflecting that the information friction resolved by a NAP is greatest for recipients about whom donors are least certain. This assumption implies that the proportional capacity shock, and therefore the allocation response, may be strongest in weak-institution settings (consistent with the governance heterogeneity in [Section 5.8](#)).

Mapping adaptive capacity to recipient merit. The variable V_i denotes donor-perceived adaptive capacity. It captures the institutional component of the IPCC vulnerability dimension, but is expressed in its favorable direction: higher V_i means greater adaptive capacity and therefore lower vulnerability. This is precisely the type of dimension that the development-aid literature treats as recipient “merit”: a recipient’s capacity to turn aid into effective projects ([Burnside and Dollar, 2000](#); [Hoeffler and Outram, 2011](#)). In our framework, the merit channel therefore corresponds to the adaptive-capacity component of IPCC vulnerability. An allocative response to higher V_i can thus be read as a response to greater perceived implementation capacity. In the language of the IPCC framework, this corresponds to a response to lower perceived vulnerability along the adaptive-capacity dimension. NAP adoption increases V_i by reducing donors’ uncertainty about planning and implementation capacity, while leaving the underlying hazard term H_i unchanged.

The CES weight. The parameter $\alpha_i = \alpha(H_i, V_i) \in (0, 1)$ governs the relative salience of adaptation in donor preferences over country i ’s aid envelope. We assume $\partial\alpha/\partial H_i \geq 0$ (donors

weight adaptation more for higher-hazard countries) and $\partial\alpha/\partial V_i \geq 0$ (donors weight adaptation more where adaptive capacity is higher, because perceived project returns rise). These two partial derivatives correspond to the two motives the empirical analysis seeks to recover: need (via hazard H_i) and merit (via adaptive capacity V_i).

3.2 Intra-country allocation (CES between C_i and O_i)

Within each recipient country, the donor first decides how to divide the country envelope between adaptation projects and other development projects.

The donor allocates an envelope $B_i \geq 0$ to country i , split into climate aid C_i and other aid O_i . The climate share $s_i \in [0, 1]$ determines

$$C_i = s_i(H_i, V_i) B_i, \quad O_i = (1 - s_i(H_i, V_i)) B_i. \quad (2)$$

Donor utility within i is a CES over (C_i, O_i) with elasticity $\sigma_c > 0$, $\sigma_c \neq 1$:

$$U_i = \left[\alpha_i C_i^{\rho_c} + (1 - \alpha_i) O_i^{\rho_c} \right]^{\frac{1}{\rho_c}}, \quad \rho_c := \frac{\sigma_c - 1}{\sigma_c}, \quad \alpha_i \equiv \alpha(H_i, V_i) \in (0, 1). \quad (3)$$

Given B_i , the intra-country problem

$$\max_{s_i \in [0, 1]} \left[\alpha_i (s_i B_i)^{\rho_c} + (1 - \alpha_i) ((1 - s_i) B_i)^{\rho_c} \right]^{\frac{1}{\rho_c}}$$

yields the standard CES share (interior solution):

$$\boxed{\frac{C_i}{O_i} = \left(\frac{\alpha_i}{1 - \alpha_i} \right)^{\sigma_c} \implies s_i^*(H_i, V_i) = \frac{\left(\frac{\alpha_i}{1 - \alpha_i} \right)^{\sigma_c}}{1 + \left(\frac{\alpha_i}{1 - \alpha_i} \right)^{\sigma_c}}.} \quad (4)$$

Substituting s_i^* into U_i and using homogeneity of degree 1 of (3):

$$U_i(B_i; H_i, V_i) = \phi_i(H_i, V_i) B_i, \quad \phi_i(H_i, V_i) := \left[\alpha_i (s_i^*)^{\rho_c} + (1 - \alpha_i) (1 - s_i^*)^{\rho_c} \right]^{\frac{1}{\rho_c}}. \quad (5)$$

Thus, H_i and V_i (through A_i) affect intra-country composition via $\alpha_i = \alpha(H_i, V_i)$, which in turn pins down ϕ_i .

3.3 Inter-country aggregation and allocation

Across countries, the donor aggregates $\{U_i\}_i$ with a CES (elasticity $\sigma_b > 0$, $\sigma_b \neq 1$):

$$U = \left(\sum_i U_i^{\rho_b} \right)^{\frac{1}{\rho_b}}, \quad \rho_b := \frac{\sigma_b - 1}{\sigma_b} \leq 1, \quad \rho_b \neq 0, \quad (6)$$

subject to the global budget

$$\sum_i B_i = B, \quad B_i \geq 0. \quad (7)$$

Using (5) and noting that $u \mapsto u^{1/\rho_b}$ is increasing on $(0, \infty)$ for $\rho_b > 0$ and decreasing for $\rho_b < 0$, maximizing U is equivalent to

$$\max_{\{B_i \geq 0\}} \sum_i (\phi_i B_i)^{\rho_b} \quad (\text{min if } \rho_b < 0) \quad \text{s.t.} \quad \sum_i B_i = B.$$

Assuming an interior solution and $\rho_b \neq 1$, first-order conditions imply

$$\boxed{B_i^* = \frac{\phi_i^{\frac{\rho_b}{1-\rho_b}}}{\sum_j \phi_j^{\frac{\rho_b}{1-\rho_b}}} B, \quad (\rho_b \neq 1).} \quad (8)$$

The associated indirect utility is

$$U^* = B \left(\sum_i \phi_i^{\frac{\rho_b}{1-\rho_b}} \right)^{\frac{1-\rho_b}{\rho_b}}. \quad (9)$$

3.4 Comparative statics

Recall: within country i , intra-country CES delivers

$$U_i = [\alpha_i C_i^{\rho_c} + (1 - \alpha_i) O_i^{\rho_c}]^{1/\rho_c}, \quad \rho_c := \frac{\sigma_c - 1}{\sigma_c}, \quad \sigma_c > 0, \quad \sigma_c \neq 1,$$

with $C_i = s_i B_i$, $O_i = (1 - s_i) B_i$, and $\alpha_i \equiv \alpha(H_i, V_i) \in (0, 1)$. The optimal intra-country share is

$$s_i^*(H_i, V_i) = \frac{\left(\frac{\alpha_i}{1 - \alpha_i} \right)^{\sigma_c}}{1 + \left(\frac{\alpha_i}{1 - \alpha_i} \right)^{\sigma_c}} \in (0, 1). \quad (10)$$

Indirect intra-country utility is linear in B_i :

$$U_i(B_i; H_i, V_i) = \phi_i(H_i, V_i) B_i, \quad \phi_i := \left[\alpha_i (s_i^*)^{\rho_c} + (1 - \alpha_i) (1 - s_i^*)^{\rho_c} \right]^{1/\rho_c}. \quad (11)$$

Across countries, inter-country CES yields

$$U = \left(\sum_i U_i^{\rho_b} \right)^{1/\rho_b}, \quad \rho_b := \frac{\sigma_b - 1}{\sigma_b} \leq 1, \quad \rho_b \neq 0, \quad (12)$$

and the optimal envelope share (interior) is

$$B_i^* = \frac{\phi_i^\eta}{\sum_j \phi_j^\eta} B, \quad \eta := \frac{\rho_b}{1 - \rho_b} = \sigma_b - 1 \quad (\rho_b \neq 1). \quad (13)$$

Notation for observables. Let $S_i := B_i$ (total aid to i), $E_i := C_i = s_i^* B_i$ (adaptation/climate aid), and $O_i := (1 - s_i^*) B_i$ (other aid). The recipient's share of global adaptation is

$$\text{Share}_i^E := \frac{E_i}{\sum_j E_j}.$$

Intra-country comparative statics

Lemma 1 (Monotonicity of the climate share). *For $\sigma_c > 0$, the optimal share s_i^* is strictly increasing in α_i :*

$$\frac{\partial s_i^*}{\partial \alpha_i} > 0.$$

If $\alpha_i = \alpha(H_i, V_i)$ with $\partial\alpha/\partial H_i \geq 0$ and $\partial\alpha/\partial V_i \geq 0$, then

$$\frac{\partial s_i^*}{\partial H_i} = \frac{\partial s_i^*}{\partial \alpha_i} \frac{\partial \alpha}{\partial H_i} \geq 0, \quad \frac{\partial s_i^*}{\partial V_i} = \frac{\partial s_i^*}{\partial \alpha_i} \frac{\partial \alpha}{\partial V_i} \geq 0.$$

Proof. From (10), write $r_i := \alpha_i/(1 - \alpha_i)$ and $q_i := r_i^{\sigma_c}$; then $s_i^* = q_i/(1 + q_i)$ and $\partial s_i^*/\partial q_i = 1/(1 + q_i)^2 > 0$, $\partial q_i/\partial r_i = \sigma_c r_i^{\sigma_c - 1} > 0$, and $\partial r_i/\partial \alpha_i = 1/(1 - \alpha_i)^2 > 0$. Chain rule yields the claim. $\square$

Corollary 1 (NAP/composition channel). *Under (1) with $\delta_i > 0$ and $x_i > 0$, switching A_i from 0 to 1 raises capacity from x_i to $V_i = (1 + \delta_i)x_i > x_i$. If $\partial\alpha/\partial V_i > 0$, then α_i rises and, by Lemma 1, s_i^* strictly increases after NAP adoption.*

Inter-country comparative statics

Lemma 2 (Envelope elasticity). *Let $\omega_i := B_i^*/B$ and $\eta := \rho_b/(1 - \rho_b)$ as in (13). Then*

$$\frac{\partial \ln B_i^*}{\partial \ln \phi_i} = \eta(1 - \omega_i).$$

Thus, the envelope response to a change in ϕ_i is larger in absolute value (i) the smaller is the recipient (ω_i low); when countries are gross substitutes in the donors aggregator ($\sigma_b > 1$, so $\eta > 0$), the response is positive and (ii) increasing in the substitutability of countries (higher ρ_b , equivalently larger σ_b).

Proof. Taking logs of (13), differentiating, and using $\sum_j \omega_j = 1$ delivers the formula. $\square$

Proposition 1 (NAP effects on levels and shares). *Suppose $A_i = 1$ increases V_i and hence α_i (i.e., $\partial\alpha/\partial V_i > 0$). Then:*

1. *Composition: $s_i^* \uparrow$ and $E_i/S_i = s_i^* \uparrow$.*

2. *Envelope*: if countries are gross substitutes in the donor's aggregator ($\sigma_b > 1$, so $\eta > 0$), then $\phi_i \uparrow$ implies $B_i^*/B = \omega_i \uparrow$ with elasticity $\eta(1 - \omega_i) > 0$.¹

3. *Outcomes*:

$$\Delta \ln E_i = \underbrace{\Delta \ln s_i^*}_{\text{composition}} + \underbrace{\Delta \ln B_i}_{\text{envelope}}, \quad \Delta \ln O_i = \Delta \ln(1 - s_i^*) + \Delta \ln B_i.$$

If the envelope effect is small on average, then $\Delta \ln E_i > 0$, $\Delta \ln S_i \simeq 0$, and $\Delta \ln O_i < 0$.

3.5 From Proposition 1 to estimation: specification and hypotheses

Proposition 1 shows how NAP adoption can affect observed aid outcomes through two distinct channels. The first is a composition channel: holding the country envelope B_i fixed, an increase in donor-perceived adaptive capacity raises the adaptation share s_i^* and therefore increases adaptation finance $E_i = s_i^* B_i$. The second is an envelope channel: if higher perceived capacity raises the donor's valuation of the recipient, the total envelope B_i may also increase. In logs, the change in adaptation finance can therefore be written as

$$\Delta \ln E_i = \Delta \ln s_i^* + \Delta \ln B_i,$$

while the corresponding changes in total and non-adaptation finance are

$$\Delta \ln S_i = \Delta \ln B_i, \quad \Delta \ln O_i = \Delta \ln(1 - s_i^*) + \Delta \ln B_i.$$

These identities discipline the empirical exercise. Estimating treatment effects for adaptation finance, total finance, non-adaptation finance, and the recipient's share of global adaptation finance allows us to distinguish whether NAP adoption mainly changes the composition of a country's aid envelope, the size of that envelope, or the country's relative position in the global allocation of adaptation finance.

Event-study specification. We implement this comparative-static logic in reduced form using a staggered event-study design. For each outcome $Y_{it} \in \{\log E_{it}, \log O_{it}, \log S_{it}\}$, and for the global adaptation-share outcome, we estimate

$$Y_{it} = \mu_i + \lambda_t + \sum_{e \neq -1} \beta_e \mathbf{1}\{t - G_i = e\} + X'_{it} \Gamma + \varepsilon_{it}, \quad (14)$$

where G_i is the year of NAP adoption, μ_i and λ_t are country and year fixed effects, and X_{it} denotes controls. The event-time coefficients β_e trace the post-adoption response of each

¹Whether NAP adoption raises ϕ_i at all is not automatic: by the envelope theorem applied to (11), $\text{sign}(\partial \phi_i / \partial \alpha_i) = \text{sign}(\alpha_i - \frac{1}{2})$, so ϕ_i is U-shaped in α_i with a minimum at $\alpha_i = \frac{1}{2}$, and a NAP-induced rise in α_i need not raise ϕ_i — which rationalizes an empirically undetectable envelope response.

outcome relative to the period immediately before adoption. Estimating the same specification across outcomes provides the reduced-form counterpart of the model’s decomposition: effects on $\log S_{it}$ capture changes in the total envelope, effects on $\log E_{it}$ combine composition and envelope responses, and effects on $\log O_{it}$ help distinguish whether higher adaptation finance crowds out other aid or is accommodated within the country envelope.

What the design identifies. The empirical design defined by Equation 14 does not estimate a coefficient on climate vulnerability. It estimates the allocation response to a dated policy signal that shifts donor-perceived adaptive capacity. In the model, persistent physical hazard is absorbed by country fixed effects, while slow-moving exposure and socio-economic sensitivity are controlled for or absorbed by fixed effects. The identifying variation therefore comes from NAP adoption, interpreted as a shift in V_i , not from changes in hazard, exposure, or socio-economic sensitivity. The estimates should therefore be read as the reduced-form allocative effect of a perceived-capacity signal, i.e., the merit channel, rather than as a response to climate need. Donor heterogeneity follows from the same logic: donors that place greater weight on planning and implementation capacity should react more strongly to NAP adoption, whereas donors whose allocation rules are driven mainly by physical need should react less, because NAP adoption does not change H_i .

Testable hypotheses. H1 (Composition versus envelope). If NAP adoption raises donor-perceived adaptive capacity, adaptation finance should increase. The cross-outcome pattern identifies the channel. A composition-dominated response predicts higher adaptation finance with little or no increase in total finance:

$$\beta_e^{(E)} > 0, \quad \beta_e^{(S)} \approx 0.$$

An envelope response predicts increases in both adaptation finance and total finance:

$$\beta_e^{(E)} > 0, \quad \beta_e^{(S)} > 0.$$

The response of non-adaptation finance distinguishes whether the increase in adaptation crowds out other aid or is accommodated within a modestly adjusted country envelope.

H2 (Global reallocation). NAP adoption should increase the recipient’s share of global adaptation finance. This outcome captures whether adopters improve their relative position in the global adaptation-finance pool after sending a capacity signal. Because the denominator is common across recipients, we interpret this outcome as evidence on reallocation within the global pool rather than as an individual treatment effect independent of other countries’ allocations.

H3 (Donor-type heterogeneity). The response to NAP adoption should be stronger among donors that condition allocation on planning and implementation capacity, such as

DAC bilateral and multilateral donors, and weaker among donors for whom such capacity is less central to allocation decisions. In the model, this corresponds to a larger $\partial\alpha/\partial V_i$ for capacity-screening donors.

The model therefore provides a clear mapping between NAP adoption and observable aid outcomes, which we test empirically using a staggered difference-in-differences design.

4 Empirical strategy

4.1 Measuring adaptation finance and the NAP signal

We build a country–year panel of adaptation-related development finance for ODA-recipient developing countries over 2009–2024. Financial data come from the OECD Creditor Reporting System (CRS), which records project-level commitments and disbursements. A project is classified as adaptation-related if adaptation is reported as either a principal or significant objective under the Rio adaptation marker. While recent research has documented inconsistencies in the coding of climate-related aid (Toetzke et al., 2022; Beaucoral, 2026), these concerns bear on whether reported flows reflect authentic adaptation content; because our object of interest is the donor allocation decision rather than the underlying adaptation effort, declared CRS commitments are precisely the relevant measure, and are well suited to capture changes in officially reported adaptation finance following NAP adoption. Moreover, we focus on commitments because they capture donor allocation decisions at the programming stage and should respond more immediately to a planning signal than disbursements.

We aggregate project-level flows to the recipient–year level. Our main outcome is log adaptation commitments. We use log adaptation disbursements as a secondary outcome, since realized flows may respond with delay as committed projects move through implementation. The remaining outcomes follow directly from the Model in Section 3. Log total commitments capture the size of the recipient’s overall development-finance envelope. Log non-adaptation commitments exclude activities marked as adaptation-related and help distinguish whether higher adaptation finance crowds out other development finance or is accommodated within the country envelope. Finally, the recipient’s share of global adaptation commitments captures whether NAP adoption improves a country’s relative position in the global adaptation-finance pool.

Treatment is defined as the first official submission of a National Adaptation Plan to the UNFCCC, recorded in NAP Central, an online platform dedicated to NAPs, and its NAP Tracking Tool that compiles up-to-date information on the submission, content and status of NAPs in developing countries². Let G_i denote country i ’s first NAP submission year, and let $G_i = \infty$ for countries that never submit a NAP during the sample period. NAPs are medium-term planning instruments introduced under the UNFCCC and used by international

²Available here: <https://napcentral.org/submitted-NAPs>

institutions as anchors for adaptation programming and finance ([World Bank, 2019](#)). In our framework, their empirical relevance is that first submission is a discrete, dated, and publicly observable signal of adaptation planning capacity.

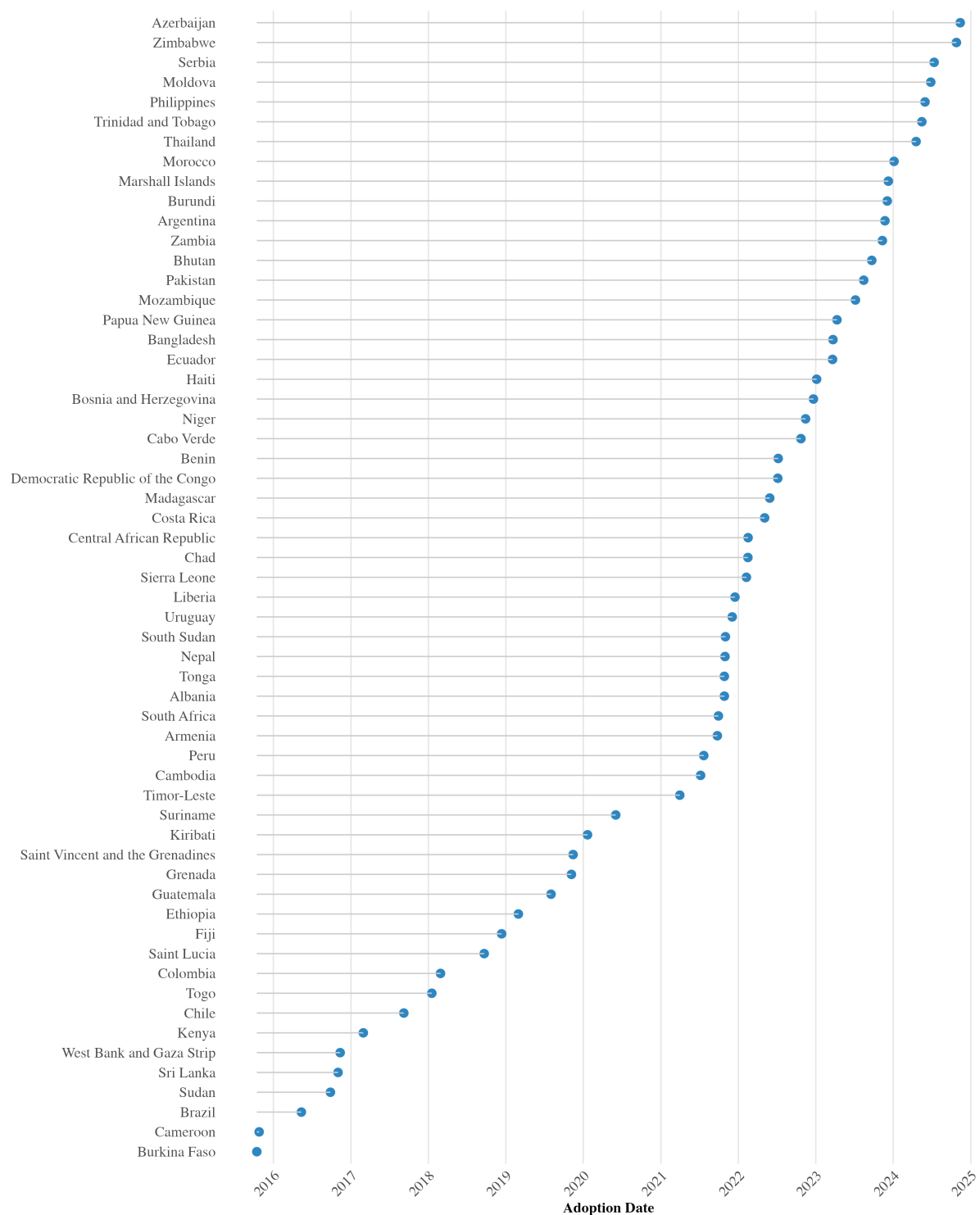


Figure 1: Adoption of National Adaptation Plans over time. *Notes:* Cumulative number of developing countries having submitted a NAP to the UNFCCC, by year of first submission. The staggered profile of adoption is the source of identifying variation in the difference-in-differences design. Source: UNFCCC NAP Central tracking tool.

Given the start of adaptation Rio-marker reporting in the CRS, the estimation panel covers 2009–2024. NAP submissions enter as treatment events from 2013 onward, and adoption is

highly staggered across countries rather than concentrated in a single global policy year. Figure 1 shows this gradual diffusion: countries submit their first NAPs in different calendar years, generating the cross-cohort timing variation on which the difference-in-differences design relies. In the main specification, the earliest estimated adoption cohort is 2015, so each treated cohort has a multi-year pre-treatment window over which pre-trends can be assessed. Additional descriptive statistics on adaptation finance, donor concentration, recipient exposure, and NAP coverage are reported in Appendix A.2.

4.2 Identification and estimation

We exploit the staggered timing of NAP adoption across countries. For each outcome Y_{it} we estimate group-time average treatment effects using the Callaway and Sant’Anna (2021) difference-in-differences estimator, which allows treatment effects to vary across adoption cohorts and calendar years and accommodates the irreversible nature of NAP submission: once a plan is submitted, it is never withdrawn, so treatment is an absorbing state. The primitive estimand is $ATT(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(\infty) \mid G_i = g]$, where $Y_{it}(g)$ is the potential outcome for a country first treated in year g and $Y_{it}(\infty)$ the potential outcome under no adoption. We summarize the $ATT(g, t)$ in two ways: a pooled effect obtained by simple aggregation (an equal-weighted average over all post-treatment cohort–time cells), and a dynamic event-study profile aggregating by event time $e = t - g$.

Our comparison group is the set of countries that never submit a NAP during the sample period. We prefer the never-treated to a not-yet-treated comparison because the not-yet-treated pool thins mechanically toward the end of the panel as adoption spreads, and because imminent adopters may already be on an upward adaptation trajectory that would contaminate the counterfactual. [à voir avec Pierre] Since never-adopters may differ systematically from adopters, we document pre-adoption comparability of the two groups in Appendix ??.

Event-time effects use a universal base period, normalized to the year immediately preceding adoption ($t = g - 1$). The identifying assumption is conditional parallel trends: absent adoption, treated and never-treated countries would have followed parallel trajectories in the relevant outcome within strata defined by the controls below. We further maintain a no-anticipation assumption, i.e., donors do not adjust commitments in expectation of a NAP before it is submitted, which we probe directly with the placebo exercise of Section 5.6.

We include two pre-treatment controls evaluated at $g - 1$: WGI Government Effectiveness, capturing baseline institutional and administrative capacity, and log population, capturing recipient scale. Because they enter at their pre-adoption level, these covariates discipline comparisons across countries with different baseline capacity and size rather than absorbing time-varying confounders; any post-adoption evolution of capacity is left in the treatment response, which is appropriate here, since the channel we study operates precisely through donors’ updated perception of adaptive capacity.

Our main specification uses the doubly robust version of the estimator (Callaway and Sant’Anna, 2021), which is consistent if either the outcome regression or the propensity-score model is correctly specified, and retains adoption cohorts with at least five treated units. Inference relies on the multiplier bootstrap (999 replications) clustered at the recipient-country level, yielding standard errors for the pooled ATT and simultaneous confidence bands for the event study.

We test the design along two dimensions. First, four checks bear on the validity of the causal interpretation, reported in Section ??: (i) pre-treatment event-study coefficients, as a graphical and formal (Wald) test of parallel trends; (ii) a placebo exercise assigning treatment two years before actual adoption; (iii) HonestDiD sensitivity bounds for departures from parallel trends (Rambachan and Roth, 2023); and (iv) a falsification test on mitigation commitments, an outcome over which NAPs carry no direct mandate. Second, we assess robustness to specification choices: we retain all adoption cohorts despite thinner cells (Section 5.3) and substitute a not-yet-treated comparison group (Section 5.4). Finally, because the outcome is measured in USD millions and adaptation commitments are zero for roughly 11% of recipient-year cells, the $\log(1 + \cdot)$ transformation is not invariant to the units of measurement (Chen and Roth, 2024); we therefore report a diagnostic across alternative transforms and units alongside a Poisson pseudo-maximum-likelihood specification in levels, which handles zeros natively and delivers a scale-invariant proportional effect (Section 5.5).

5 Results

We estimate the causal effect of NAP adoption on five theoretically motivated climate finance outcomes using the Callaway and Sant’Anna (2021) staggered difference-in-differences estimator. All specifications use the never-treated group as the comparison and control for WGI Government Effectiveness and log population evaluated at pre-treatment values (base period $g - 1$); controls enter through doubly-robust estimation in the main specification and through regression adjustment when thin cohorts are retained (Section 4). Two specifications are reported: the main specification drops cohorts with fewer than five treated units and uses bootstrap standard errors (999 replications); the robustness specification retains all cohorts and uses asymptotic standard errors.

Throughout, we read the estimates through the framework of Section 3. The design holds the need components of climate risk fixed — hazard absorbed by country fixed effects, human exposure proxied (crudely) by population — and conditions on baseline capacity, so the coefficients capture the allocative response to the NAP information shock, which we interpret as the NAP-induced increment in donors’ perceived adaptive capacity V_i , not a response to any change in recipient need.

5.1 Average treatment effects

Table 2: Effect of NAP adoption on climate finance: simple ATT across outcomes (cohorts ≥ 5 units, DR + multiplier-bootstrap SE)

	log(Adaptation commitments)	Adaptation share (% of global)	log(Total commitments)	log(Non-adaptation commitments)	log(Adaptation disbursements)
ATT	0.2937**	0.3185**	0.1010	0.0705	0.1421
SE	(0.1245)	(0.1403)	(0.0637)	(0.0642)	(0.1167)
<i>t</i> -statistic	2.360	2.269	1.586	1.098	1.218
Mean (pre-treat., USD M)	101.1	0.77 pp	1850.7	1749.6	68.7
Effect (USD M)	34.5	—	196.7	127.7	10.5
Observations	2,014	2,014	2,014	2,014	2,014
Countries	126	126	126	126	126
Pre-trend χ^2	2.022	2.349	9.529	10.228	3.484
Pre-trend <i>p</i> -value	0.732	0.672	0.049	0.037	0.480
Estimator	CS (2021)	CS (2021)	CS (2021)	CS (2021)	CS (2021)
Control group	Never-treated	Never-treated	Never-treated	Never-treated	Never-treated
Bootstrap SE	Yes (multiplier, 999 reps)	Yes (multiplier, 999 reps)	Yes (multiplier, 999 reps)	Yes (multiplier, 999 reps)	Yes (multiplier, 999 reps)

Notes: CS (2021) doubly-robust; WGI gov. effectiveness + log population; never-treated control group; multiplier-bootstrap SE (999 replications). Pre-trend test: Wald χ^2 on joint significance of pre-treatment event-time ATTs. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: OECD CRS (Rio adaptation markers); UNFCCC NAP Central.

Table 2 reports the simple (equal-weighted, calendar-time-pooled) ATT for each outcome under the main specification.

Hypothesis H1 — adaptation commitments and envelope expansion. The ATT on log adaptation commitments is 0.294 ($SE = 0.125$, $p < 0.05$), corresponding to a $\exp(0.294) - 1 \approx 34\%$ increase. Total commitments show a small, statistically insignificant ATT (0.101, $SE = 0.064$), and non-adaptation commitments (0.071, $SE = 0.064$) are likewise indistinguishable from zero. Together, these estimates point to a mechanism in which NAP adoption *reallocates* finance toward adaptation rather than expanding the overall envelope: the sharp rise in adaptation flows is not matched by a significant increase in total commitments. In terms of the model’s decomposition $\Delta \ln E_{it} = \Delta \ln s_{it}^* + \Delta \ln B_{it}$ (composition plus scale; Proposition 1), this is the composition channel: the within-country adaptation share s_{it}^* rises while the total envelope B_{it} is essentially unchanged, so the adaptation increase reflects a shift in the within-country allocation rather than a pure scale effect. The flat non-adaptation estimate ($\beta_e^{(O)} \approx 0$) is consistent with this reading: the rise in adaptation is accommodated within a roughly constant envelope — the composition-with-offset case (H1.1), in which a small envelope uptick absorbs the mechanical decline in non-climate aid that the pure composition channel (H1) would otherwise imply ($\beta_e^{(O)} < 0$). Read through the model, this is the capacity channel at work: a higher perceived V_i lifts the weight donors place on adaptation within the adopter’s envelope (α_i , hence s_{it}^*), while the flat total indicates that any envelope response (the ϕ_i channel) is too small to detect. One caveat applies to this reading: the formal pre-trend tests for total and non-adaptation commitments reject at conventional levels (Table 2), so the flatness of these two secondary outcomes rests on a weaker identification footing than the adaptation and share results; we document this in Section 5.2.

Hypothesis H2 — reallocation of global adaptation finance. The ATT on the adaptation share outcome is 0.319 percentage points ($SE = 0.140$, $p < 0.05$), providing support for the reallocation mechanism.³ The share effect is quantitatively meaningful: in a distribution where many recipients hold shares below one percentage point, a gain of 0.3 pp represents a substantial relative improvement in positioning. Taken together with the H1 results, the evidence is consistent with a composition-and-reallocation mechanism: NAP adoption shifts the within-country allocation toward adaptation *and* shifts the global allocation of adaptation finance toward adopters, at the expense of non-adopters, without a significant change in the total envelope — the pattern expected when a broadly fixed global pool of adaptation finance is redirected toward recipients whose perceived capacity to absorb it has just been revised upward.

Disbursements. The ATT on log adaptation disbursements is 0.142 ($SE = 0.117$), not significant at conventional levels. The attenuation relative to commitments is consistent with well-documented implementation lags: projects committed after NAP adoption typically enter multi-year disbursement schedules, so commitment-year signals may take several additional years to materialize in disbursed flows.

³The share outcome is constructed as $share_{it} = Y_{it} / \sum_j Y_{jt} \times 100$. This cross-sectional denominator induces a SUTVA violation: a treated unit's gain mechanically reduces the potential outcome of untreated units. The DiD estimand therefore captures a net reallocation effect rather than an uncontaminated individual treatment effect, and should be read as descriptive consistency with the reallocation mechanism rather than a clean causal estimate.

5.2 Dynamic treatment effects

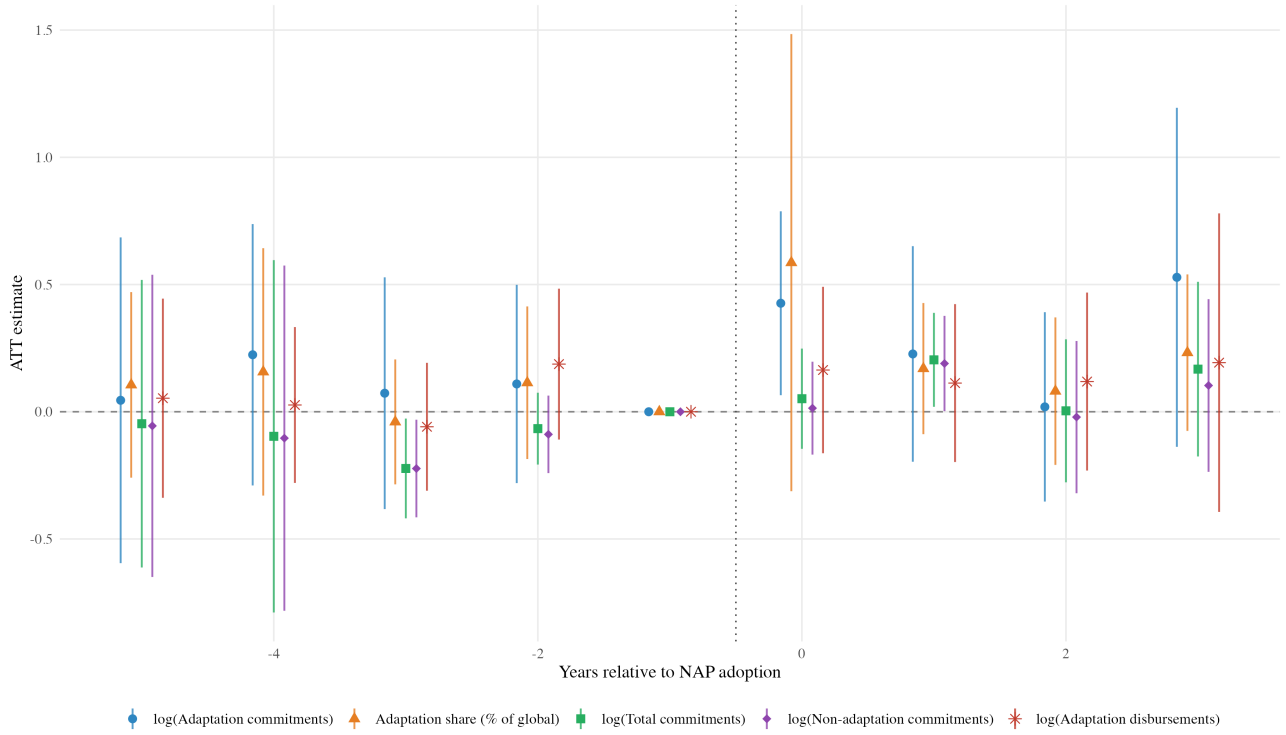


Figure 2: Dynamic ATT of NAP adoption across five outcomes. CS(2021); cohorts ≥ 5 treated units; bootstrap 95% simultaneous CI; controls: WGI Government Effectiveness, log population; base period $g - 1$. Source: authors' estimations on OECD CRS (Rio adaptation markers) and UNFCCC NAP Central data.

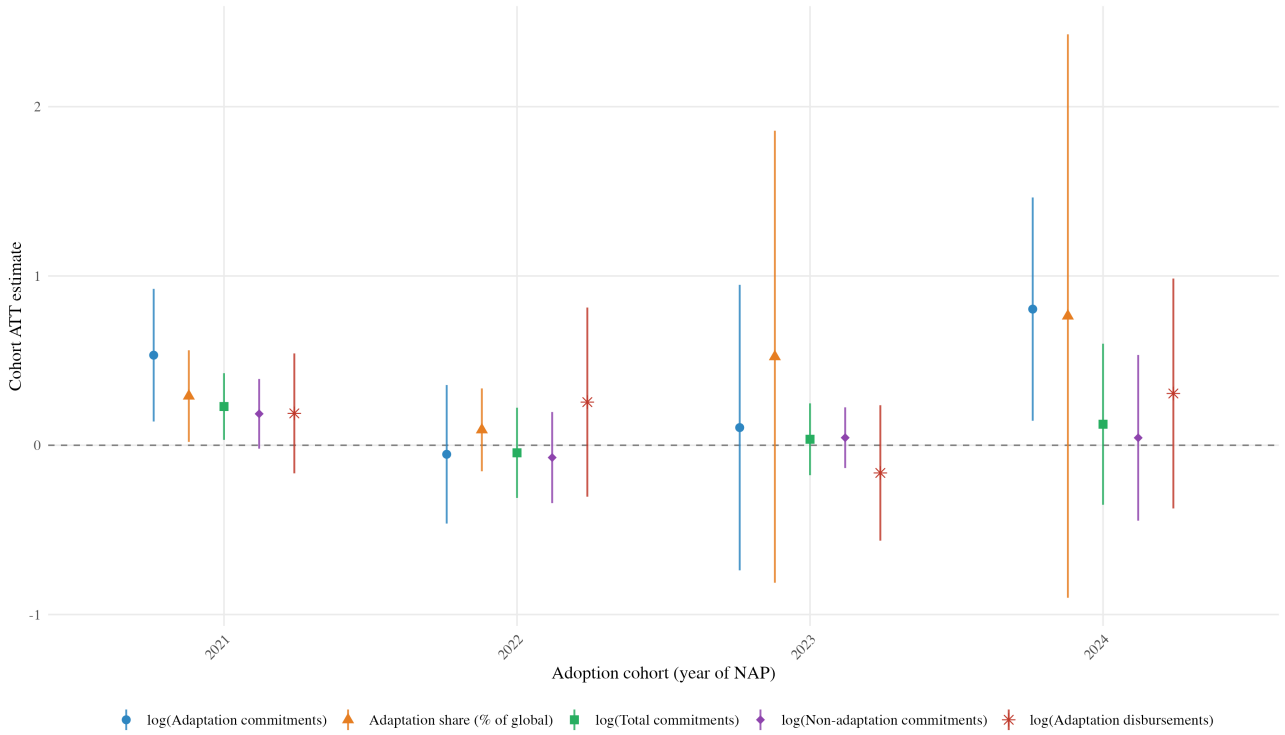


Figure 3: ATT by adoption cohort across five outcomes. CS(2021); cohorts ≥ 5 treated units; bootstrap SE; controls: WGI Government Effectiveness, log population. Source: authors' estimations on OECD CRS (Rio adaptation markers) and UNFCCC NAP Central data.

Figure 2 displays the combined event-study profiles for all five outcomes. For the adaptation, share, and disbursement outcomes, pre-trend coefficients are small and concentrated near zero, and the formal Wald tests do not reject parallel pre-treatment trends (adaptation: $\chi^2 = 2.022$, $p = 0.732$; share: $\chi^2 = 2.349$, $p = 0.672$; Table 2). The picture is less clean for the two aggregate outcomes: the Wald test rejects for total commitments ($p = 0.049$) and non-adaptation commitments ($p = 0.037$). This matters for the composition-with-offset reading (H1.1), which leans on these two outcomes being flat: if their counterfactual trends are not parallel, the “flat total, flat non-adaptation” pattern cannot be given a sharp causal interpretation, and we treat it as suggestive rather than established. The core conclusion survives this caveat because the channel evidence rests primarily on the adaptation and share outcomes — whose pre-trends pass both the graphical and formal checks — while the total and non-adaptation estimates play a secondary, corroborating role. Post-adoption, adaptation commitments rise progressively rather than jumping on impact, consistent with donor programming cycles converting an updated assessment of adaptive capacity into commitments over several budget rounds. Effects are modest in early post-adoption periods, which reflects both the small number of treatment-eligible cohorts in the main specification (the ≥ 5 units filter restricts estimation to a subset of adoption years) and the relatively short post-treatment window available in the sample.

Figure 3 decomposes the pooled ATT by adoption cohort. The 2021 cohort — the largest treated cohort in the sample — is the primary driver of the estimated positive effect on adapta-

tion commitments. Smaller adjacent cohorts produce noisier individual estimates that average down in the pooled ATT, explaining the moderate aggregate effect size.

5.3 Robustness: retaining thin cohorts

Table 3: Effect of NAP adoption on climate finance: simple ATT across outcomes (all cohorts retained, asymptotic SE)

	log(Adaptation commitments)	Adaptation share (% of global)	log(Total commitments)	log(Non-adaptation commitments)	log(Adaptation disbursements)
ATT	0.1182	0.1723*	0.0004	-0.0130	-0.0528
SE	(0.1271)	(0.0913)	(0.2116)	(0.2092)	(0.0952)
<i>t</i> -statistic	0.930	1.889	0.002	-0.062	-0.554
Mean (pre-treat., USD M)	95.6	0.79 pp	1812.5	1716.9	65.9
Effect (USD M)	12.0	—	0.8	-22.1	-3.4
Observations	2,302	2,302	2,302	2,302	2,302
Countries	144	144	144	144	144
Pre-trend χ^2	4.733	3.694	3.337	3.676	4.654
Pre-trend <i>p</i> -value	0.316	0.449	0.503	0.452	0.325
Estimator	CS (2021)	CS (2021)	CS (2021)	CS (2021)	CS (2021)
Control group	Never-treated	Never-treated	Never-treated	Never-treated	Never-treated
Bootstrap SE	No (analytical SE)	No (analytical SE)	No (analytical SE)	No (analytical SE)	No (analytical SE)

Notes: CS (2021) outcome regression; WGI gov. effectiveness + log population; never-treated control group; analytical (influence-function) SE. Pre-trend test: Wald χ^2 on joint significance of pre-treatment event-time ATTs. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: OECD CRS (Rio adaptation markers); UNFCCC NAP Central.

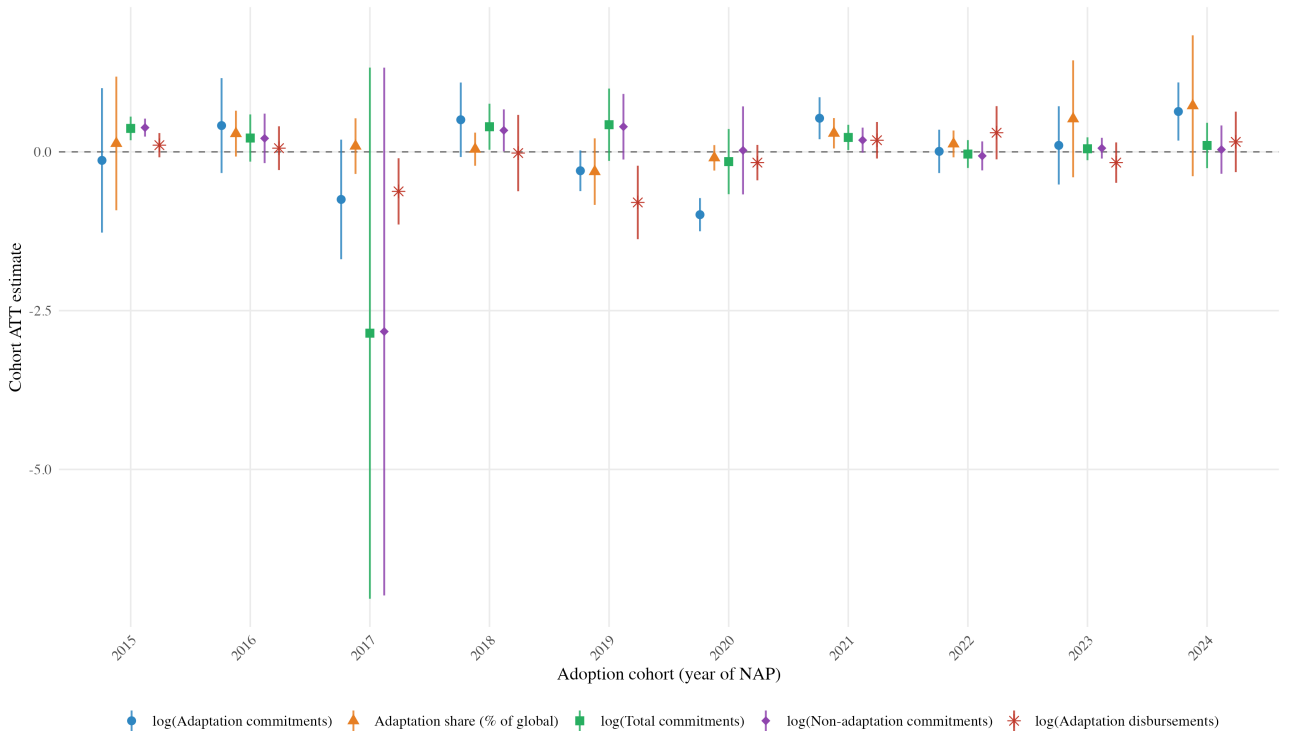


Figure 4: ATT by adoption cohort across five outcomes, all cohorts retained. CS(2021); asymptotic SE; controls: WGI Government Effectiveness, log population. Source: authors' estimations on OECD CRS (Rio adaptation markers) and UNFCCC NAP Central data.

As a sensitivity check, we re-estimate all models retaining cohorts with fewer than five treated units, using asymptotic standard errors. This extends the effective treatment window and

brings in early-adopter cohorts excluded from the main specification. A placebo test assigning treatment two years before actual adoption is presented in Section 5.6.

When early (thin) cohorts are included, the estimates shift in a predictable way. Early adopter cohorts (pre-2018) show elevated non-adaptation aid effects, which we attribute primarily to a data reliability artefact: Rio marker classification in early CRS vintages is less consistent, and some non-adaptation flows may have been incorrectly tagged in early years, inflating estimates for that outcome. For later cohorts (post-2019) the pattern aligns closely with the main specification — adaptation commitments rise, non-adaptation is flat or slightly negative — and the 2021 cohort results remain the most precisely estimated. The share effect remains directionally consistent, though only marginally significant in this specification: the ATT is 0.172 percentage points ($SE = 0.091$, $p < 0.10$). When the thin early cohorts are retained, the reallocation pattern is therefore more sensitive to cohort inclusion than the main specification, where it is significant at the 5% level. These results reinforce the core composition interpretation while indicating that early-cohort estimates should be treated with caution due to Rio marker data quality concerns in early CRS vintages.

5.4 Robustness: not-yet-treated control group

As a further sensitivity check, we re-estimate the main specification using the not-yet-treated rather than the never-treated comparison group, holding everything else fixed (cohorts ≥ 5 treated units, doubly-robust estimation, multiplier-bootstrap standard errors, controls WGI Government Effectiveness and log population; $N = 2,014$, 126 countries). Under this comparison, countries that adopt a NAP in a later year serve as controls for earlier-adopting cohorts until their own adoption, while never-treated units remain in the control pool throughout. A stable estimate under this alternative comparison set shows that the result is not an artefact of how the never-treated group is constructed.

Table 4: Effect of NAP adoption on climate finance: simple ATT across outcomes (not-yet-treated controls, multiplier-bootstrap SE)

	log(Adaptation commitments)	Adaptation share (% of global)	log(Total commitments)	log(Non-adaptation commitments)	log(Adaptation disbursements)
ATT	0.2800**	0.3209**	0.0907	0.0607	0.1339
SE	(0.1254)	(0.1434)	(0.0624)	(0.0610)	(0.1136)
<i>t</i> -statistic	2.233	2.238	1.455	0.995	1.179
Mean (pre-treat., USD M)	101.1	0.77 pp	1850.7	1749.6	68.7
Effect (USD M)	32.6	—	175.7	109.5	9.8
Observations	2,014	2,014	2,014	2,014	2,014
Countries	126	126	126	126	126
Pre-trend χ^2	2.102	2.813	8.463	9.026	3.692
Pre-trend <i>p</i> -value	0.717	0.590	0.076	0.060	0.449
Estimator	CS (2021)	CS (2021)	CS (2021)	CS (2021)	CS (2021)
Control group	Not-yet-treated	Not-yet-treated	Not-yet-treated	Not-yet-treated	Not-yet-treated
Bootstrap SE	Yes (multiplier, 999 reps)	Yes (multiplier, 999 reps)	Yes (multiplier, 999 reps)	Yes (multiplier, 999 reps)	Yes (multiplier, 999 reps)

Notes: CS (2021) doubly-robust; WGI gov. effectiveness + log population; not-yet-treated control group; multiplier-bootstrap SE (999 replications). Pre-trend test: Wald χ^2 on joint significance of pre-treatment event-time ATTs. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: OECD CRS (Rio adaptation markers); UNFCCC NAP Central.

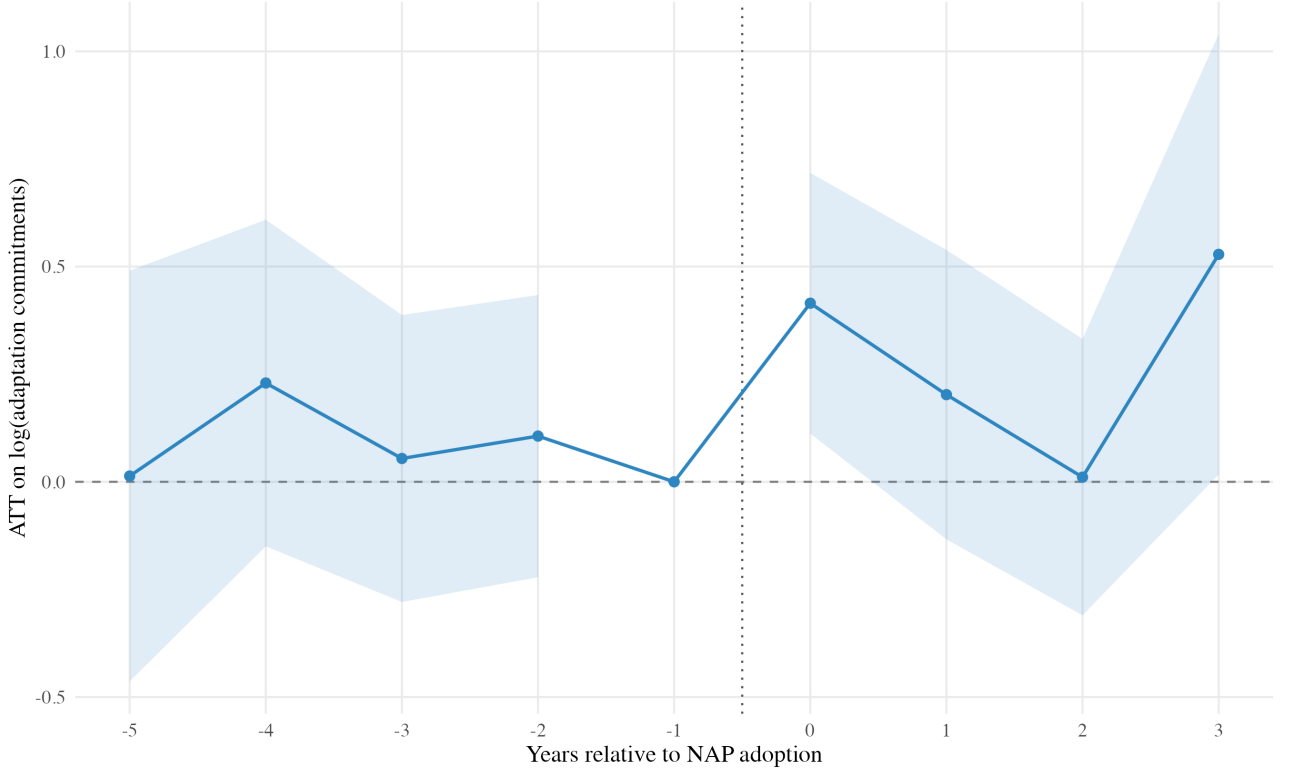


Figure 5: Dynamic ATT of NAP adoption across five outcomes, not-yet-treated control group. *Notes:* CS (2021) DR; cohorts ≥ 5 treated units; multiplier-bootstrap 95% CI; not-yet-treated control; controls: WGI Government Effectiveness, log population; base period $g - 1$. Flat pre-trends and a positive on-impact effect indicate the dynamics do not depend on the choice of control group. Source: authors' estimations on OECD CRS (Rio adaptation markers) and UNFCCC NAP Central data.

Table 4 and Figure 5 report the results. The ATT on log adaptation commitments is 0.280 ($SE = 0.125$, $p < 0.05$), essentially unchanged from the main estimate of 0.294 ($SE = 0.125$) in Table 2, and the adaptation share is likewise stable at 0.321 ($SE = 0.143$, $p < 0.05$) against 0.319 in the main specification. The secondary outcomes remain statistically indistinguishable from zero and close to their main-specification values: total commitments 0.091 ($SE = 0.062$), non-adaptation commitments 0.061 ($SE = 0.061$), and disbursements 0.134 ($SE = 0.114$). The pre-trend tests reproduce the pattern of the main specification (Section 5.2): adaptation and the share pass comfortably ($\chi^2 = 2.102$, $p = 0.717$; $\chi^2 = 2.813$, $p = 0.590$), while total and non-adaptation commitments remain borderline ($p = 0.076$; $p = 0.060$). We read this consistency as confirming that the weaker identification footing of the two aggregate outcomes is a feature of the data rather than of the never-treated comparison. The event study (Figure 5) shows flat, insignificant pre-trends and a positive on-impact effect. Read through the model of Section 3, the composition channel — a rise in the within-country adaptation share s_{it}^* at a roughly fixed envelope, consistent with H1 and H1.1 — survives the change in comparison set intact. Two further checks reported in the appendix corroborate the design: an alternative heterogeneity-robust estimator that reproduces the headline effect (Appendix A.3), and a Goodman-Bacon

decomposition showing that the forbidden comparisons carry negligible weight in the two-way fixed-effects estimate (Appendix A.4).

5.5 Robustness: outcome units and the treatment of zeros

As an additional check on the headline result for adaptation commitments, we examine its dependence on the units in which the outcome is measured and on the way exact zeros are handled. The baseline outcome is the `log1p` of adaptation commitments measured in USD *millions*. With 230 of the 2,014 observations (11.4%) recording exact zeros, alongside many sub-million flows, the `log1p` transform is sensitive to the units of measurement: the implicit offset (“+1”) means “+1 million” at the millions scale but “+1 dollar” in raw USD — a well-known property of log-like transforms in the presence of zeros (Chen and Roth, 2024). We test whether this property, rather than the underlying response, drives the estimate.

Table 5: Sensitivity of the headline ATT to outcome units and the treatment of zeros

Specification	ATT	SE	t	Pre-trend χ^2	p	N
<code>log1p</code> (USD millions)	0.2937**	(0.1173)	2.503	2.022	0.732	2,014
<code>log1p</code> (raw USD)	0.2772	(0.1749)	1.585	3.506	0.477	2,014
<code>asinh</code> (USD millions)	0.2954**	(0.1251)	2.362	2.487	0.647	2,014
<code>asinh</code> (raw USD)	0.2766	(0.1792)	1.544	3.455	0.485	2,014
Pure log, zeros dropped	0.3307**	(0.1396)	2.369	2.839	0.585	1,784
PPML, levels (Poisson)	0.3707***	(0.1246)	2.975	—	—	2,014

Notes: Outcome is adaptation commitments. Specs 1–5 are CS (2021): specs 1–4 doubly-robust with multiplier-bootstrap SE (999 replications); spec 5 outcome-regression with bootstrap SE on the positive-commitment subsample. The final row (PPML, levels) is the Sun & Abraham (2021) interaction-weighted Poisson pseudo-maximum-likelihood estimator with country and year fixed effects, clustered by country; it estimates `commitments` in levels. Never-treated control group; WGI gov. effectiveness + log population controls. `commitments` in USD millions, raw USD = $\times 10^6$; the pure-log / drop-zeros and Poisson-in-levels specifications are invariant to the units of the outcome, and Poisson additionally retains zero-valued observations; its coefficient is a proportional (semi-elasticity) effect. Pre-trend test: Wald χ^2 on joint significance of pre-treatment event-time ATTs. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: OECD CRS (Rio adaptation markers); UNFCCC NAP Central.

Table 5 reports the results. The baseline `log1p`-millions estimate, reproducing the main specification, is 0.294 ($SE = 0.117$, $p < 0.05$). Re-expressing commitments in raw USD leaves the *point estimate* almost unchanged at 0.277 ($SE = 0.176$) but inflates the standard error so that the estimate is no longer significant ($t = 1.59$). The inverse hyperbolic sine reproduces the identical pattern (millions: 0.295, $SE = 0.125$, $p < 0.05$; raw USD: 0.277, $SE = 0.180$, ns), confirming that the sensitivity is generic to zero-accommodating transforms rather than specific to `log1p`. The cleanest check is the pure log estimated on the positive-commitment subsample, which drops the 230 zeros: a change of units is then an additive constant that differences out, so this intensive-margin estimate is *invariant to units*. It is 0.331 ($SE = 0.140$, $p < 0.05$, $t = 2.37$), if anything slightly larger than the baseline. The most complete check

is a Poisson pseudo-maximum-likelihood specification in levels, estimated with the interaction-weighted event-study approach of [Sun and Abraham \(2021\)](#) (country and year fixed effects, standard errors clustered by country). This is the only specification that is *both* invariant to the units of the outcome *and* retains the 230 zero-valued observations, and its coefficient is a scale-invariant proportional effect ([Chen and Roth, 2024](#)). The PPML estimate is 0.371 ($SE = 0.125$, $p < 0.01$; $\exp(0.371) - 1 \approx 45\%$), the largest and most precisely estimated of the six specifications.

The effect *size* is therefore stable across every specification, ranging only over 0.277–0.331; what moves is the *precision* of the zero-inclusive transforms, and that movement is an artefact of how the 230 zeros are encoded rather than evidence against the result. The intensive-margin response is significant and depends on neither units nor any offset. The PPML specification reinforces this reading on a broader basis: estimated on the full sample, with the zeros retained and free of any units dependence, it corroborates the result on the extensive plus intensive margin rather than the intensive margin alone, and does so as the most precisely estimated of the six specifications. Read transparently — in the spirit of the HonestDiD discussion of [Section 5.9](#) — the headline effect rests on a robust, unit-invariant increase that holds whether or not the zeros are included, and we flag the units-dependence of the `log1p` and `asinh` precision as a known caveat rather than a threat to the conclusion.

5.6 Placebo test

As an additional check on the parallel trends assumption, we conduct a placebo test by assigning a fictitious treatment date two years before each country’s actual NAP adoption. We restrict the sample for treated units to pre-actual-treatment observations only: a treated country enters the placebo panel only for years $t < G_i$, with cohort year set to $G_i - 2$. Never-treated countries contribute their full panel. Because all observations for treated units in this placebo panel predate their actual NAP adoption, any statistically significant ATT at the placebo date would signal pre-existing differential trends rather than a causal effect of NAP adoption, directly undermining the parallel trends assumption. Estimation uses regression adjustment with bootstrap standard errors: the propensity-score step of the doubly-robust estimator is infeasible in the shifted placebo panel, whose truncated cohorts generate degenerate cells.

Table 6: Placebo test: treatment date assigned 2 years before actual NAP adoption

Placebo (treatment shifted -2 years)	
ATT	-0.0192
SE	(0.1503)
t -statistic	-0.128
Mean (pre-treat., USD M)	87.3
Effect (USD M)	-1.7
Observations	1,910
Countries	126
Pre-trend χ^2	0.915
Pre-trend p -value	0.822
Estimator	CS (2021)
Control group	Never-treated
Bootstrap SE	Yes

Notes: Treatment date assigned 2 years before actual NAP adoption. Only pre-actual-treatment observations for treated units are included. A statistically insignificant ATT supports the parallel trends assumption. CS (2021) regression adjustment; bootstrap SE (999 reps); never-treated control; controls: WGI gov. effectiveness + log population. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: OECD CRS (Rio adaptation markers); UNFCCC NAP Central.

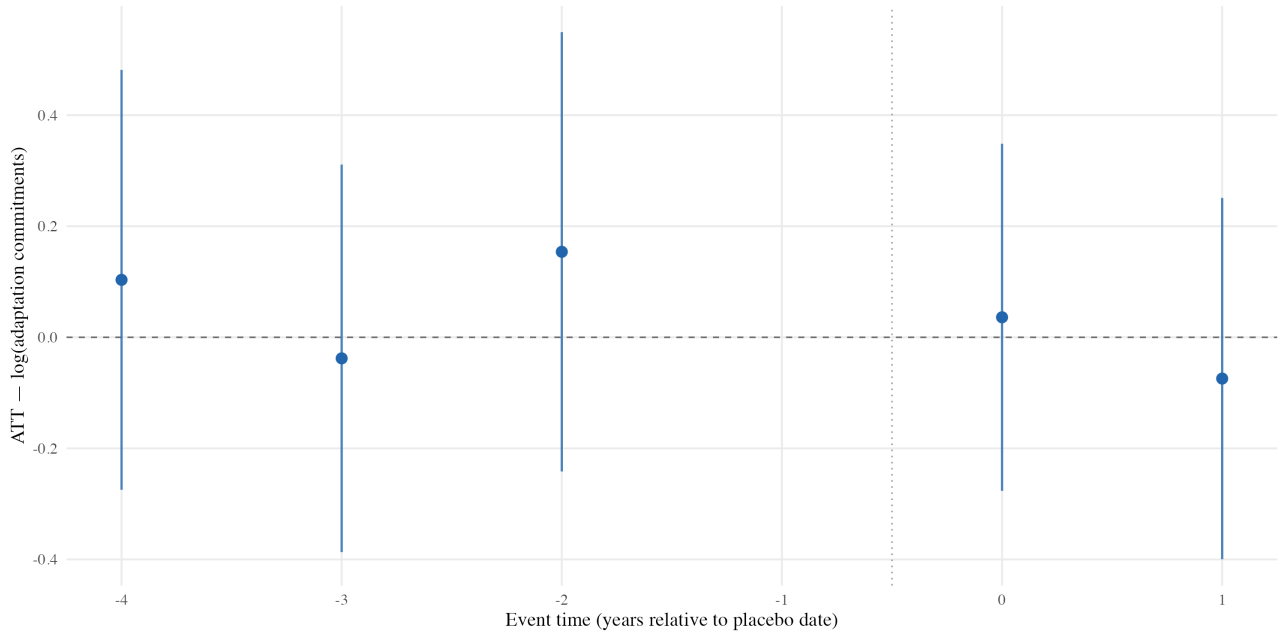


Figure 6: Placebo test: event-study ATT with treatment date shifted -2 years. *Notes:* CS (2021) regression adjustment; bootstrap 95% CI; never-treated control; only pre-actual-treatment observations for treated units; controls: WGI Government Effectiveness, log population. A flat profile centred on zero supports the parallel trends assumption. Sources: OECD CRS; UNFCCC NAP Central.

Table 6 and Figure 6 report the results. The placebo ATT is -0.019 ($SE = 0.150$, $t = -0.13$), statistically indistinguishable from zero. The pre-trend test yields $\chi^2 = 0.915$ ($p = 0.822$), confirming a flat pre-placebo profile. The point estimate is essentially zero, with no

sign of an upward pre-trend that would inflate the main estimates. Crucially, there is no evidence of a positive spurious effect at the fake treatment date: adaptation finance was not already diverging upward two years before actual NAP adoption. This directly supports the parallel trends assumption underlying the main specification. The timing is also informative about the mechanism. If donors were responding to a gradual improvement in recipient capacity or engagement that merely culminated in a NAP, finance should already drift upward before submission; instead, the response coincides with the submission event itself, consistent with the plan operating as a discrete information shock to donors’ perception of adaptive capacity.

5.7 Mitigation falsification test

As a further falsification test, we estimate the same CS (2021) model using log mitigation commitments as the outcome. NAPs are adaptation-specific policy instruments with no direct mandate over mitigation finance. If the positive effect on adaptation commitments were driven by a broader climate-engagement signal — countries that adopt NAPs also becoming more active on mitigation and thereby attracting all forms of climate aid — one would expect a similarly sized positive coefficient on mitigation flows. A null result, by contrast, supports the interpretation that donors specifically respond to the adaptation content of NAPs rather than to an undifferentiated climate-policy posture.

Table 7: Falsification test: effect of NAP adoption on log(mitigation commitments)

	log(Mitigation commitments)
ATT	0.1728
SE	(0.1522)
<i>t</i> -statistic	1.136
Mean (pre-treat., USD M)	134.3
Effect (USD M)	25.3
Observations	2,014
Countries	126
Pre-trend χ^2	3.556
Pre-trend <i>p</i> -value	0.469
Estimator	CS (2021)
Control group	Never-treated
Bootstrap SE	Yes (multiplier, 999 reps)

Notes: NAPs are adaptation-focused instruments; a null result here supports the interpretation that the positive adaptation effect is not driven by broad climate-policy signalling. CS (2021) doubly-robust; multiplier-bootstrap SE (999 replications); never-treated control; cohorts ≥ 5 units; controls: WGI gov. effectiveness + log population. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: OECD CRS (Rio adaptation markers); UNFCCC NAP Central.

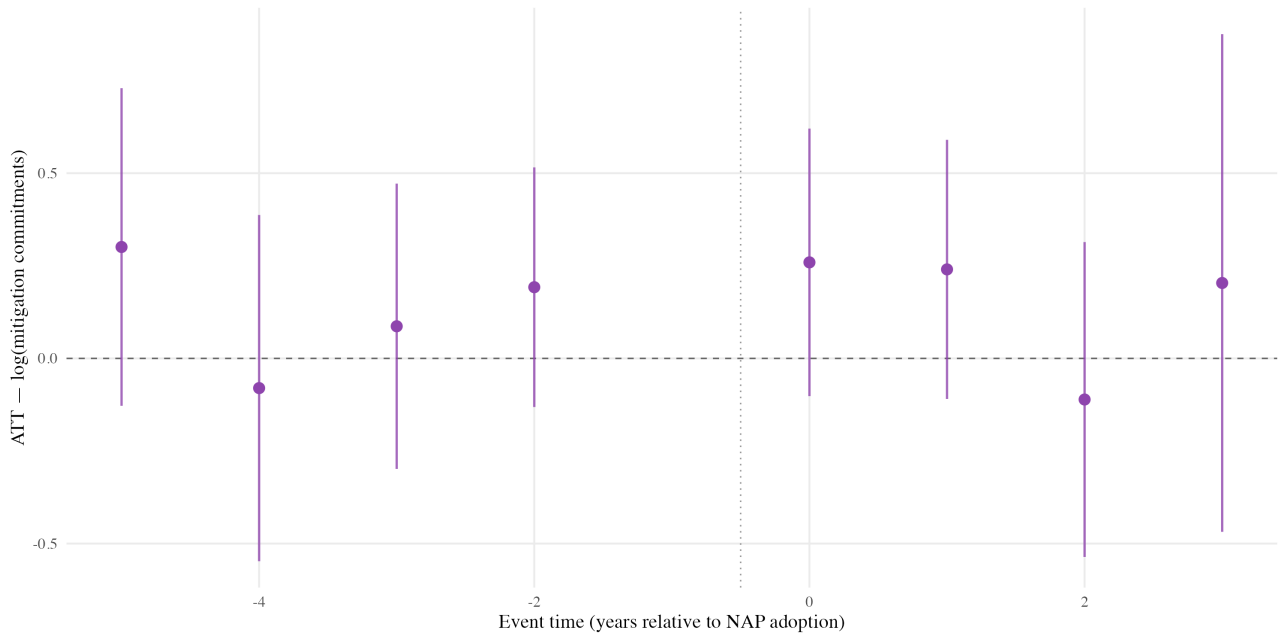


Figure 7: Falsification test: dynamic ATT on log(mitigation commitments) following NAP adoption. CS (2021); never-treated control; controls: WGI Government Effectiveness, log population. A null result supports specificity of the adaptation effect. Source: authors’ estimations on OECD CRS (Rio mitigation markers) and UNFCCC NAP Central data.

Table 7 and Figure 7 report the results. The estimated ATT on mitigation commitments is 0.173 ($SE = 0.152$, $t = 1.14$), statistically indistinguishable from zero, and the pre-trend test does not reject ($\chi^2 = 3.556$, $p = 0.469$). The point estimate is positive but imprecise, so the falsification evidence is a contrast in statistical strength rather than a sharp zero: the adaptation effect is significant and builds dynamically, while the mitigation profile shows no significant effect in any post-treatment period. This asymmetry supports the causal interpretation of the main results and sharpens it: the information donors extract from a NAP appears to concern a recipient’s capacity to plan and implement *adaptation* — the V_i channel of Section 3 — rather than a general climate-policy posture that would raise all climate flows.

5.8 Heterogeneity analyses

To shed light on the mechanisms behind the average effects, we examine heterogeneity along four dimensions: donor type, recipient governance capacity, LDC status, and World Bank income group. Donor-type results decompose the outcome on the full estimation sample and therefore use the main specification (cohorts ≥ 5 treated units, doubly-robust estimator, bootstrap SE, never-treated control). The remaining analyses split the sample; our protocol applies doubly-robust estimation only when a subgroup contains at least 40 treated countries, and since every governance, LDC, and income subgroup falls below that threshold, all sample-split estimates use regression adjustment with analytical standard errors. These three splits also adopt a not-yet-treated comparison group: in the smallest subsamples the never-treated control pool is too

thin to serve as a reliable comparison under the current estimator, which flags it as inadequate for these splits, so later-adopting countries supply the additional pre-adoption comparisons that the never-treated group alone cannot.⁴ Where subsamples are small, low statistical power — rather than the absence of an effect — is the most plausible explanation for non-significant estimates.

Donor type. We decompose adaptation commitments by donor category — DAC bilaterals, multilaterals, and other donors — and estimate the ATT separately for each flow. Table 8 reports the results. DAC bilateral flows rise by 0.308 log points ($SE = 0.147$, $p < 0.05$, $\approx 36\%$) and multilateral flows by 0.551 log points ($SE = 0.250$, $p < 0.05$, $\approx 74\%$) following NAP adoption. By contrast, the ATT for other (non-traditional) donors is 0.012 ($SE = 0.127$), statistically indistinguishable from zero. This pattern is consistent with hypothesis H3: NAP adoption functions as a readiness signal that primarily activates capacity-oriented public donors. DAC bilaterals and multilateral development institutions — which systematically weight institutional capacity in allocation decisions — respond strongly, while non-traditional donors do not. Because a NAP is informative about adaptive capacity (V_i) but not about hazard, an allocation rule that places little weight on perceived capacity leaves the signal with nothing to update — precisely the muted response the model predicts for these donors.

Table 8: Effect of NAP adoption on climate finance: simple ATT across outcomes (heterogeneity by donor type; cohorts ≥ 5 units, DR estimator)

	DAC bilateral	Multilateral	Other donors
ATT	0.3075**	0.5513**	0.0116
SE	(0.1470)	(0.2502)	(0.1273)
t -statistic	2.091	2.203	0.091
Mean (pre-treat., USD M)	90.5	10.3	0.3
Effect (USD M)	32.6	7.6	0.0
Observations	2,014	2,014	2,014
Countries	126	126	126
Pre-trend χ^2	2.300	3.181	9.875
Pre-trend p -value	0.681	0.528	0.043
Estimator	CS (2021)	CS (2021)	CS (2021)
Control group	Never-treated	Never-treated	Never-treated
Bootstrap SE	Yes (multiplier, 999 reps)	Yes (multiplier, 999 reps)	Yes (multiplier, 999 reps)

Notes: CS (2021) doubly-robust; WGI gov. effectiveness + log population; never-treated control group; multiplier-bootstrap SE (999 replications). Pre-trend test: Wald χ^2 on joint significance of pre-treatment event-time ATTs. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: OECD CRS (Rio adaptation markers); UNFCCC NAP Central.

⁴The donor-type decomposition retains the never-treated control because it is estimated on the full sample rather than a split.

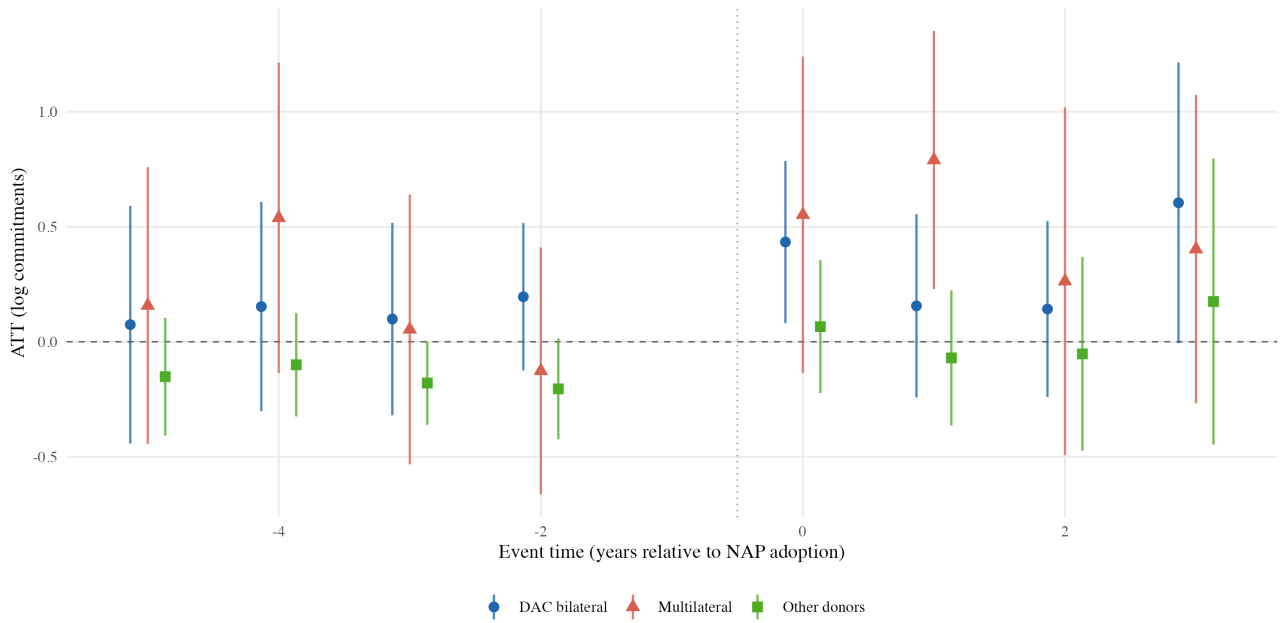


Figure 8: Dynamic ATT by donor type. CS (2021) DR; cohorts ≥ 5 treated units; bootstrap 95% CI; never-treated control; controls: WGI Government Effectiveness, log population. Source: authors' estimations on OECD CRS (Rio adaptation markers) and UNFCCC NAP Central data.

Governance capacity. We split the sample at the median pre-treatment WGI Government Effectiveness score. Table 9 reports the results. The ATT is 0.297 log points ($SE = 0.118$, $p < 0.05$) for *low*-governance countries and 0.293 ($SE = 0.233$, ns) for high-governance countries. The effect is thus statistically concentrated in countries with below-median institutional quality, consistent with a credentialing interpretation: NAP adoption conveys a stronger incremental signal precisely where baseline institutions are weaker, since donors face greater uncertainty about implementation capacity in these contexts. Because baseline Government Effectiveness is already conditioned on, the split suggests where the signal's increment matters most: the same plan shifts donor beliefs about capacity furthest where the prior is weakest. A note of caution is warranted, however: the formal pre-trend test for the low-governance subgroup yields a Wald χ^2 of 9.996 ($p = 0.040$), indicating some deviation from parallel pre-treatment trends. This raises the possibility that low-governance adopters were already on an upward trajectory in adaptation receipts before NAP submission, which could upward-bias the ATT for this group. The estimate should therefore be interpreted with care.

Table 9: Heterogeneity by governance level: ATT on log(adaptation commitments)

	High governance	Low governance
ATT	0.2931	0.2965**
SE	(0.2329)	(0.1179)
<i>t</i> -statistic	1.258	2.515
Mean (pre-treat., USD M)	90.5	110.9
Effect (USD M)	30.8	38.3
Observations	992	1,022
Countries	62	64
Pre-trend χ^2	2.351	9.996
Pre-trend <i>p</i> -value	0.671	0.040
Estimator	CS (2021) regression adjustment	CS (2021) regression adjustment
Control group	Not-yet-treated	Not-yet-treated
Bootstrap SE	No (analytical SE)	No (analytical SE)

Notes: Heterogeneity by governance level: ATT on log(adaptation commitments). CS (2021); analytical SE; not-yet-treated control; median WGI GE split at baseline; cohorts ≥ 5 treated units. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: OECD CRS (Rio adaptation markers); UNFCCC NAP Central.

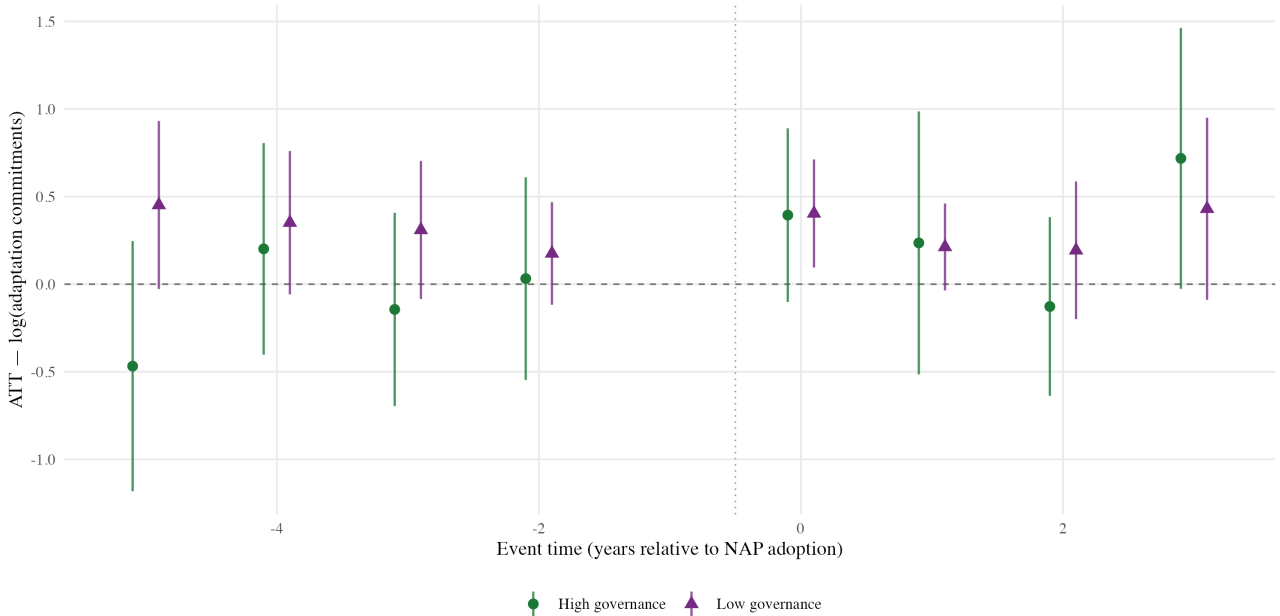


Figure 9: Dynamic ATT by governance level (median WGI GE split at baseline). CS (2021) regression adjustment; analytical SE; never-treated control; controls: WGI Government Effectiveness, log population. Source: authors' estimations on OECD CRS (Rio adaptation markers), UNFCCC NAP Central, and Worldwide Governance Indicators data.

LDC status. Table 10 reports ATTs for Least Developed Countries (LDCs) and non-LDC recipients separately. Neither subgroup yields a statistically significant estimate: the ATT is 0.271 ($SE = 0.324$) for LDCs and 0.167 ($SE = 0.188$) for non-LDCs. These null results should be interpreted with caution. The LDC subsample comprises only 17 countries, a size

at which the estimator has low statistical power and analytical standard errors are wide. The point estimates are economically meaningful and broadly consistent with the main-sample ATT, suggesting that the absence of significance reflects insufficient precision rather than a genuine absence of effect. Further replication with larger samples or longer time horizons would be needed to resolve this question.

Table 10: Heterogeneity by LDC status: ATT on log(adaptation commitments)

	LDC	Non-LDC
ATT	0.2708	0.1673
SE	(0.3238)	(0.1883)
<i>t</i> -statistic	0.836	0.888
Mean (pre-treat., USD M)	131.2	82.4
Effect (USD M)	40.8	15.0
Observations	255	1,759
Countries	17	124
Pre-trend χ^2	3.193	2.282
Pre-trend <i>p</i> -value	0.526	0.684
Estimator	CS (2021) regression adjustment	
Control group	Not-yet-treated	
Bootstrap SE	No (analytical SE)	

Notes: Heterogeneity by LDC status: ATT on log(adaptation commitments). CS (2021); analytical SE; not-yet-treated control; cohorts ≥ 5 treated units. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: OECD CRS (Rio adaptation markers); UNFCCC NAP Central.

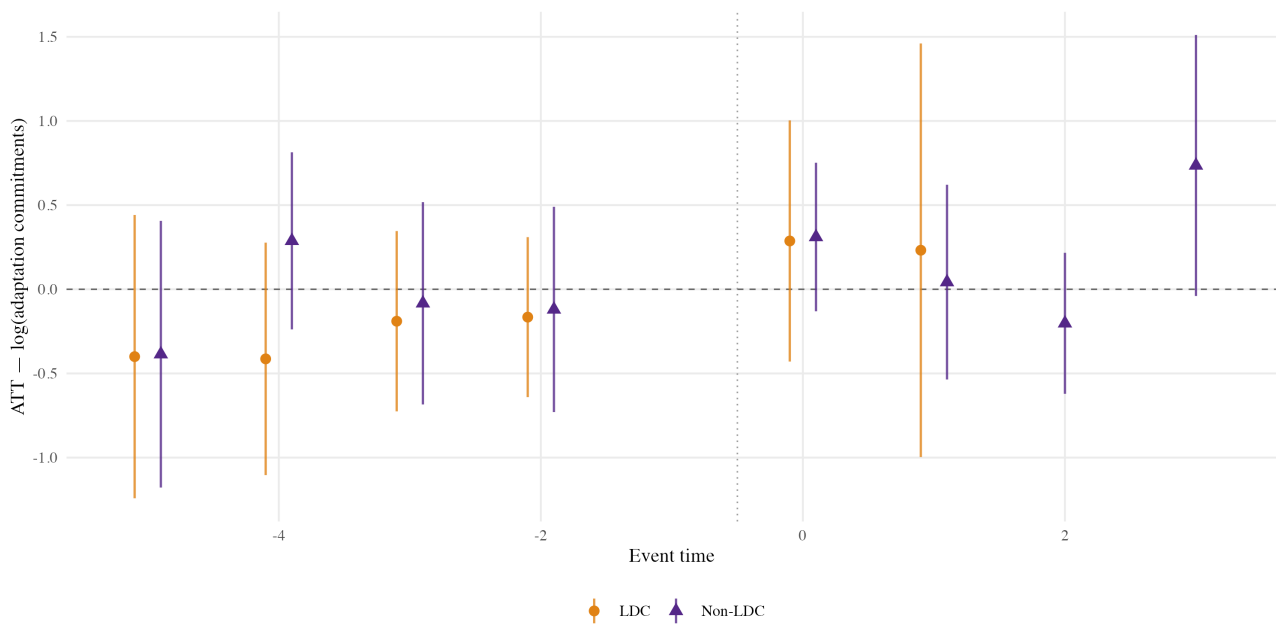


Figure 10: Dynamic ATT by LDC status. CS (2021) regression adjustment; analytical SE; never-treated control; controls: WGI Government Effectiveness, log population. Source: authors' estimations on OECD CRS (Rio adaptation markers), UNFCCC NAP Central, and UN LDC classification data.

Income group. Table 11 disaggregates recipients by World Bank income classification. Among the three groups with sufficient treated cohorts, the ATT is marginally significant for low-income countries (0.417 log points, $SE = 0.221$, $p < 0.10$; $e^{0.417} - 1 \approx 52\%$) and significant at the 5% level for lower-middle-income countries (0.319, $SE = 0.162$, $p < 0.05$, $\approx 38\%$), while upper-middle-income countries yield a positive but imprecise estimate (0.297, $SE = 0.287$, ns). The low-income group, however, exhibits a clear pre-trend concern: the Wald test rejects at the 5% level ($\chi^2 = 11.135$, $p = 0.025$), so the low-income estimate rests on a weaker identification footing and should not be read as cleanly causal. As with the LDC split, limited subgroup sizes constrain statistical power; the imprecise upper-middle-income estimate is more plausibly a reflection of that constraint than of a structural absence of effect. Lower-middle-income adopters — a group of 39 countries whose estimate is both significant and free of pre-trend concerns — emerge as the clearest beneficiaries in this decomposition.

Table 11: Heterogeneity by World Bank income group: ATT on log(adaptation commitments)

	Low income	Lower middle income	Upper middle income
ATT	0.4169*	0.3189**	0.2970
SE	(0.2206)	(0.1623)	(0.2866)
<i>t</i> -statistic	1.890	1.965	1.036
Mean (pre-treat., USD M)	94.2	166.6	60.7
Effect (USD M)	48.7	62.6	21.0
Observations	270	624	688
Countries	17	39	43
Pre-trend χ^2	11.135	1.191	4.213
Pre-trend <i>p</i> -value	0.025	0.880	0.378
Estimator	CS (2021) regression adjustment	CS (2021) regression adjustment	CS (2021) regression adjustment
Control group	Not-yet-treated	Not-yet-treated	Not-yet-treated
Bootstrap SE	No (analytical SE)	No (analytical SE)	No (analytical SE)

Notes: Heterogeneity by World Bank income group: ATT on log(adaptation commitments). CS (2021); analytical SE; not-yet-treated control; cohorts ≥ 5 treated units. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: OECD CRS (Rio adaptation markers); UNFCCC NAP Central.

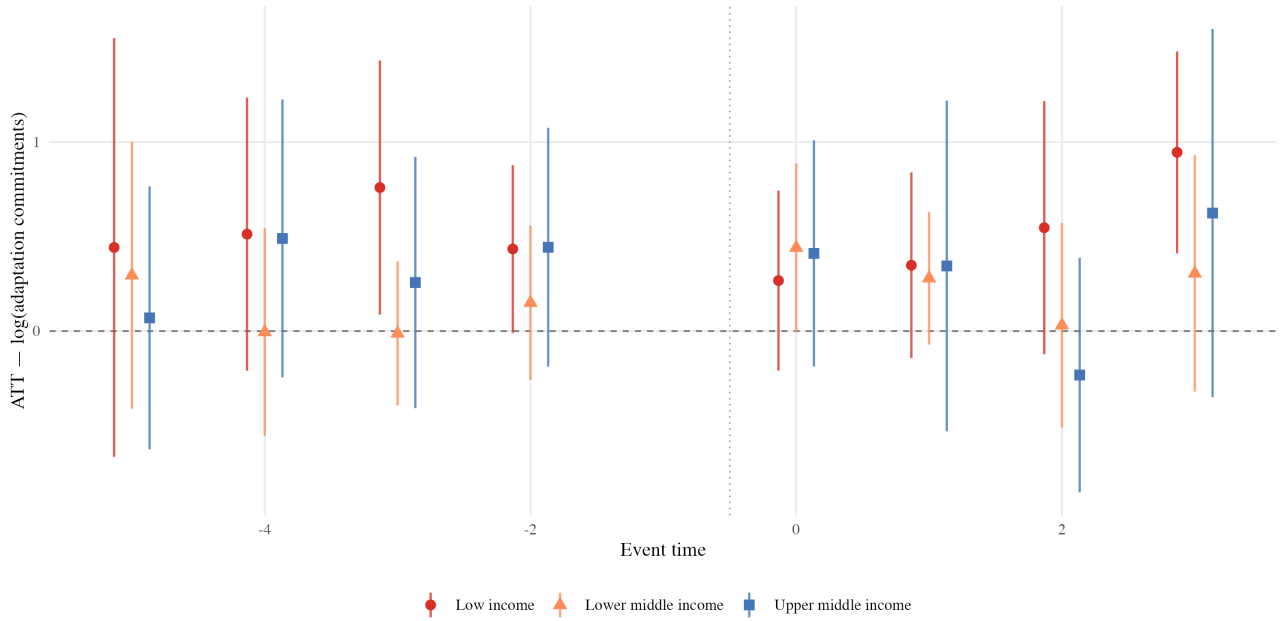


Figure 11: Dynamic ATT by World Bank income group. CS (2021) regression adjustment; analytical SE; never-treated control; controls: WGI Government Effectiveness, log population. Source: authors' estimations on OECD CRS (Rio adaptation markers), UNFCCC NAP Central, and World Bank income classification data.

5.9 Sensitivity to parallel trends violations

Table 12: HonestDiD robust 95% CI under relative magnitude restriction (Rambachan and Roth 2023)

$\bar{M}$	log(Adapt.)	Share (pp)	log(Total)	log(Non-adapt.)	log(Disb.)
$\bar{M} = 0.00$	[0.063, 0.556]	[0.069, 0.480]	[-0.031, 0.255]	[-0.062, 0.219]	[-0.070, 0.369]
$\bar{M} = 0.25$	[-0.134, 0.728]	[-0.124, 0.669]	[-0.208, 0.433]	[-0.236, 0.401]	[-0.215, 0.554]
$\bar{M} = 0.50$	[-0.449, 1.033]	[-0.426, 0.971]	[-0.482, 0.710]	[-0.506, 0.674]	[-0.436, 0.798]
$\bar{M} = 0.75$	[-0.779, 1.367]	[-0.745, 1.290]	[-0.765, 0.992]	[-0.787, 0.955]	[-0.694, 1.061]
$\bar{M} = 1.00$	[-1.119, 1.702]	[-1.068, 1.613]	[-1.053, 1.278]	[-1.071, 1.236]	[-0.961, 1.332]
$\bar{M} = 1.25$	[-1.459, 2.042]	[-1.395, 1.941]	[-1.339, 1.456]	[-1.355, 1.448]	[-1.232, 1.608]
$\bar{M} = 1.50$	[-1.804, 2.382]	[-1.722, 2.096]	[-1.456, 1.456]	[-1.448, 1.448]	[-1.508, 1.884]
$\bar{M} = 1.75$	[-2.144, 2.535]	[-2.050, 2.096]	[-1.456, 1.456]	[-1.448, 1.448]	[-1.784, 2.160]
$\bar{M} = 2.00$	[-2.489, 2.535]	[-2.096, 2.096]	[-1.456, 1.456]	[-1.448, 1.448]	[-2.060, 2.259]
Breakdown $\bar{M}$	0.25	0.25	0.00	0.00	0.00

Notes: Equal-weighted average ATT across post-treatment periods. $\bar{M} = 0$: standard 95% CI. $\bar{M} = 1$: violation up to max observed pre-trend. CS (2021) outcome regression; never-treated control group; analytical SE. Source: OECD CRS (Rio adaptation markers); UNFCCC NAP Central.

Following Rambachan and Roth (2023), we assess robustness under the relative magnitude restriction $\Delta_{\bar{M}}^{RM}$: the post-treatment bias path may deviate from zero by at most $\bar{M}$ times the maximum observed pre-treatment deviation. Table 12 reports robust 95% CIs for the equal-weighted average ATT across post-treatment periods at each $\bar{M}$ level. At $\bar{M} = 0$ the

entries recover the standard 95% CIs; increasing $\bar{M}$ widens the bounds to accommodate growing violations of parallel trends.

Both headline results break down at $\bar{M} = 0.25$. For adaptation commitments, the robust CI widens from $[0.063, 0.556]$ at $\bar{M} = 0$ to $[-0.134, 0.728]$ at $\bar{M} = 0.25$; for the global adaptation share, from $[0.069, 0.480]$ to $[-0.124, 0.669]$. The positive effects are therefore guaranteed only if post-treatment violations of parallel trends stay below 25% of the maximum observed pre-trend deviation. The intervals for the remaining outcomes include zero already at $\bar{M} = 0$: for log total commitments, $[-0.031, 0.255]$, consistent with the insignificant pooled ATT in Table 2 and with the flat-total conclusion (whose pre-trends are, in any case, the weakest; Section 5.2). We interpret the sensitivity analysis transparently: with a short post-treatment window, even moderate allowances for parallel-trend violations widen the bounds quickly, and caution is warranted against over-interpreting either the adaptation level estimate or the share estimate as definitively causal. The evidence is most credibly described as consistent with the reallocation hypothesis, with both headline results robust only to moderate violations of parallel trends.

6 Conclusion

This paper combines a parsimonious CES model with modern staggered DiD estimators to study whether National Adaptation Plans (NAPs) attract adaptation finance. In the model, NAPs raise a country's effective capacity, which increases the donor's climate salience and thus the *climate share* of its envelope. Adaptation receipts decompose as $\Delta \ln E_{it} = \Delta \ln s_{it}^* + \Delta \ln B_{it}$, so impacts operate through a *composition* channel ($s_{it}^* \uparrow$) and/or a *scale* channel ($B_{it} \uparrow$). Consistent with our estimates, a “composition with offset” case can arise: $s_{it}^* \uparrow$ while B_{it} and non-climate aid remain statistically flat because a small envelope uptick offsets the mechanical decline in other aid.

Empirically, the estimates point to a dual mechanism. Adaptation commitments rise after NAP adoption ($\approx 34\%$, $p < 0.05$), while total commitments are statistically flat and non-adaptation commitments are flat. The rise in adaptation flows reflects a reallocation within a roughly constant envelope, consistent with a composition channel ($s_{it}^* \uparrow$) rather than a scale effect. Simultaneously, a country's share of global adaptation finance increases by approximately 0.3 percentage points ($p < 0.05$, robust across cohort specifications), consistent with a reallocation of global adaptation resources toward adopters.

Heterogeneity analyses sharpen the mechanism. The donor decomposition supports H3: DAC bilateral ($\approx 36\%$, $p < 0.05$) and multilateral donors ($\approx 74\%$, $p < 0.05$) drive the effect, while non-traditional donors show no response. The governance split reveals a significant effect concentrated in below-median governance countries ($p < 0.05$), consistent with a credentialing interpretation, though a pre-trend deviation in that subgroup ($p = 0.040$) means the estimate should be treated with caution. The LDC and non-LDC subsamples yield statistically

insignificant estimates, most plausibly due to the small LDC sample (17 countries) rather than a structural absence of effect. Among income groups, low-income and lower-middle-income adopters show positive effects ($\approx 38\text{--}52\%$), while upper-middle-income countries do not; the weaker precision reflects limited subgroup sample sizes and, for low-income countries, a pre-trend concern that rejects parallel trends at the 5% level ($p = 0.025$).

Policy implications. For recipients, the NAP process yields returns beyond planning: by signaling institutional readiness, it attracts a larger share of global adaptation finance, predominantly from DAC bilaterals and multilaterals. The concentration of effects in lower-governance countries suggests NAPs are most effective as financial catalysts where they close the largest credibility gap. For donors, explicitly using NAP adoption as an allocation criterion can improve targeting efficiency; coordination mechanisms should guard against excessive concentration in a few high-profile adopters at the expense of non-adopters. Because the mechanism rewards the adaptive-capacity (merit) dimension of climate risk, it risks steering finance away from high-need, low-capacity non-adopters—countries high on hazard, exposure, and sensitivity but with weak institutions—precisely the need–merit trade-off our framework makes visible.

Limitations and next steps. Our results use commitments and Rio-marker classification; both are imperfect proxies for realized, adaptation-relevant impact. The DiD design isolates average responses but cannot fully separate composition from scale without auxiliary information on S_{it} ; moreover, formal pre-trend tests reject for total and non-adaptation commitments (Section 5.2), so the flatness of the aggregate envelope is suggestive rather than established. HonestDiD sensitivity analysis shows that both the adaptation commitments result and the share result are guaranteed only if post-treatment violations stay below 25% of the maximum observed pre-trend deviation ($\bar{M} = 0.25$); given the short post-treatment window, this warrants caution against over-causal interpretation of both headline estimates. Future work could: (i) examine disbursements and execution lags; (ii) track project-level channels (sectors, instruments) to test model margins; (iii) study duration and decay of effects beyond five years; and (iv) explore interactions with NDCs and climate funds. Finally, by construction the design identifies the adaptive-capacity (merit) channel: the need components of climate risk—hazard, exposure, and sensitivity—are held fixed by fixed effects and controls rather than estimated, so our evidence speaks to how capacity signals move finance, not to whether donors reward need. A residual interpretive caveat is that NAP adoption may also reveal planning and need information to donors; the mitigation falsification rules out a generic climate-engagement credential, but not an adaptation-specific need-revelation channel, so we read the estimate as operating primarily—though not necessarily exclusively—through the adaptive-capacity channel.

Takeaway. *NAPs pay* — not by equally expanding all envelopes, but by signaling readiness to the donors who respond to it. The effect concentrates in DAC and multilateral flows,

is largest for lower-governance adopters, and manifests as a shift of each adopter's envelope toward adaptation and a reallocation of global adaptation resources toward adopters, within a broadly constant total envelope.

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A Appendix

A.1 Additional derivative expressions

For $r_i := \alpha_i / (1 - \alpha_i)$ and $q_i := r_i^{\sigma_c}$,

$$s_i^* = \frac{q_i}{1 + q_i}, \quad \frac{\partial s_i^*}{\partial \alpha_i} = \frac{q_i}{(1 + q_i)^2} \cdot \frac{\sigma_c}{\alpha_i(1 - \alpha_i)}.$$

Hence, by chain rule,

$$\frac{\partial s_i^*}{\partial w} = \frac{q_i}{(1 + q_i)^2} \cdot \frac{\sigma_c}{\alpha_i(1 - \alpha_i)} \cdot \frac{\partial \alpha}{\partial w}, \quad w \in \{H_i, V_i\}.$$

From (13), log-differentiating with B fixed,

$$\frac{\partial \ln B_i^*}{\partial \ln \phi_i} = \frac{\partial \ln \omega_i}{\partial \ln \phi_i} = \eta(1 - \omega_i),$$

since $B_i^* = \omega_i B$ with B fixed. The envelope response of recipient i *relative to* the share-weighted average $\sum_j \omega_j \eta(1 - \omega_j) = \eta(1 - \bar{\omega})$ is therefore

$$\eta(1 - \omega_i) - \eta(1 - \bar{\omega}) = \eta(\bar{\omega} - \omega_i),$$

where $\bar{\omega} := \sum_j \omega_j^2$ is the Herfindahl of envelopes.

A.2 Additional data description

This appendix provides additional descriptive information on the construction and coverage of the data used in the empirical analysis. It documents the evolution of adaptation-related finance, the concentration of flows across donors and recipients, and the timing of National Adaptation Plan (NAP) adoption across countries.

A.2.1 Adaptation finance over time

Figure A1 reports annual adaptation-related commitments and disbursements over the estimation period. Commitments increase substantially after the introduction of Rio adaptation reporting, while disbursements evolve more smoothly and tend to lag commitments. This pattern motivates our focus on commitments as the main allocation outcome and the use of disbursements as a secondary outcome.

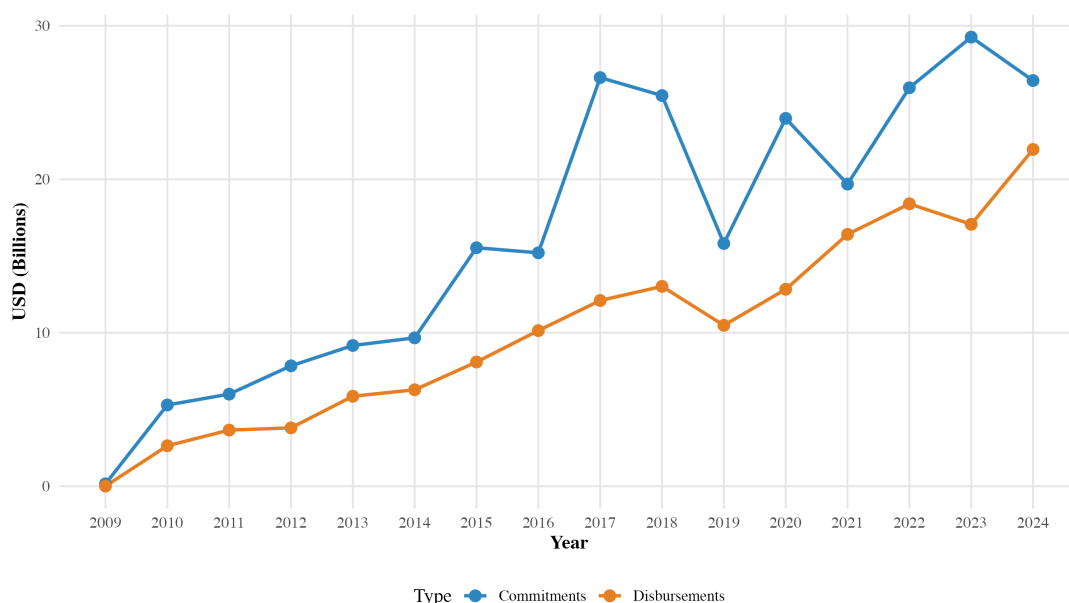


Figure A1: Evolution of adaptation-related commitments and disbursements, 2009–2024

Notes: Annual totals of adaptation-related commitments and disbursements in constant USD billion across all recipient countries. Each point is the sum of all CRS activities marked as adaptation-relevant under the Rio adaptation marker, either as a principal or significant objective. Source: OECD Creditor Reporting System.

Table A1: Annual summary statistics of adaptation-related finance

Year	Total commitments (USD bn)	Total disbursements (USD bn)	Recipients	Mean commitments (USD bn)	Median commitments (USD bn)	SD commitments (USD bn)
2009	0.16	0.01	10	0.02	0.00	0.04
2010	5.29	2.64	142	0.04	0.01	0.07
2011	6.00	3.66	142	0.04	0.01	0.07
2012	7.85	3.80	141	0.05	0.01	0.10
2013	9.17	5.86	144	0.05	0.01	0.08
2014	9.67	6.28	141	0.04	0.01	0.09
2015	15.54	8.09	141	0.07	0.02	0.13
2016	15.21	10.14	143	0.06	0.01	0.11
2017	26.63	12.10	144	0.09	0.01	0.21
2018	25.46	13.02	141	0.08	0.01	0.22
2019	15.82	10.48	142	0.05	0.01	0.12
2020	23.97	12.83	141	0.07	0.01	0.24
2021	19.69	16.41	141	0.06	0.00	0.15
2022	25.96	18.40	140	0.07	0.00	0.19
2023	29.26	17.07	140	0.09	0.01	0.22
2024	26.43	21.94	140	0.08	0.01	0.18

Notes: Adaptation-related finance is identified using the Rio adaptation marker in the OECD Creditor Reporting System. Totals sum all adaptation-related flows in each year. The number of recipients counts distinct recipient countries with adaptation-marked flows. Mean, median, and standard deviation are computed across recipient–donor-type cells within each year. The 2009 row reflects the partial first year of adaptation marker reporting. Source: OECD CRS.

A.2.2 Donor and recipient concentration

Adaptation-related finance is concentrated among a limited set of donors and recipients. Figure A2 ranks the largest providers over the sample period, while Figure A3 maps cumulative commitments by recipient country. These patterns motivate the donor-type heterogeneity analysis in the main text but are not used for identification.

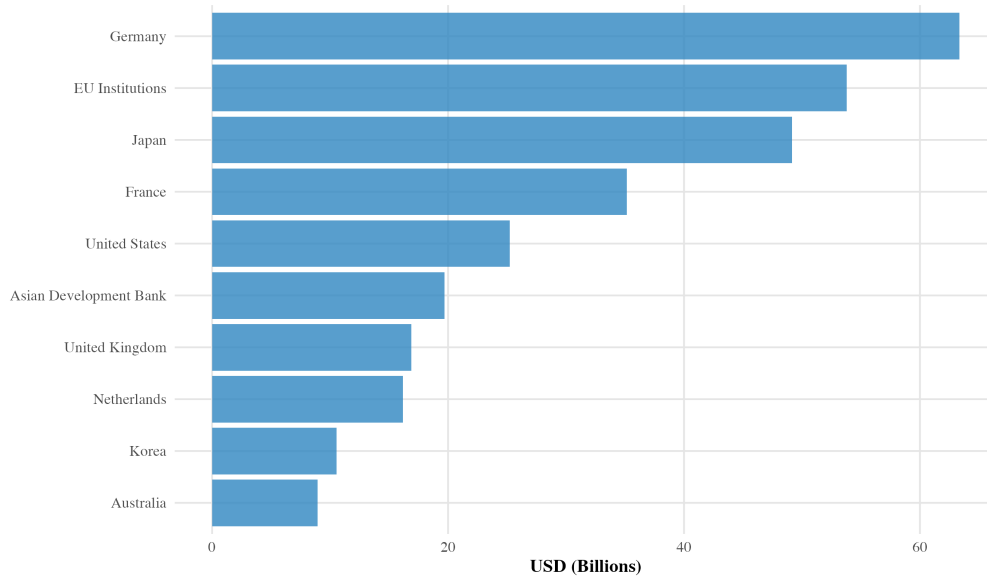


Figure A2: Top donors for adaptation-related finance

Notes: The figure reports the ten largest providers of adaptation-related finance, ranked by cumulative commitments in constant USD billion over 2009–2024. Bars sum all CRS activities marked as adaptation-relevant under the Rio adaptation marker, either as a principal or significant objective. Source: OECD Creditor Reporting System.

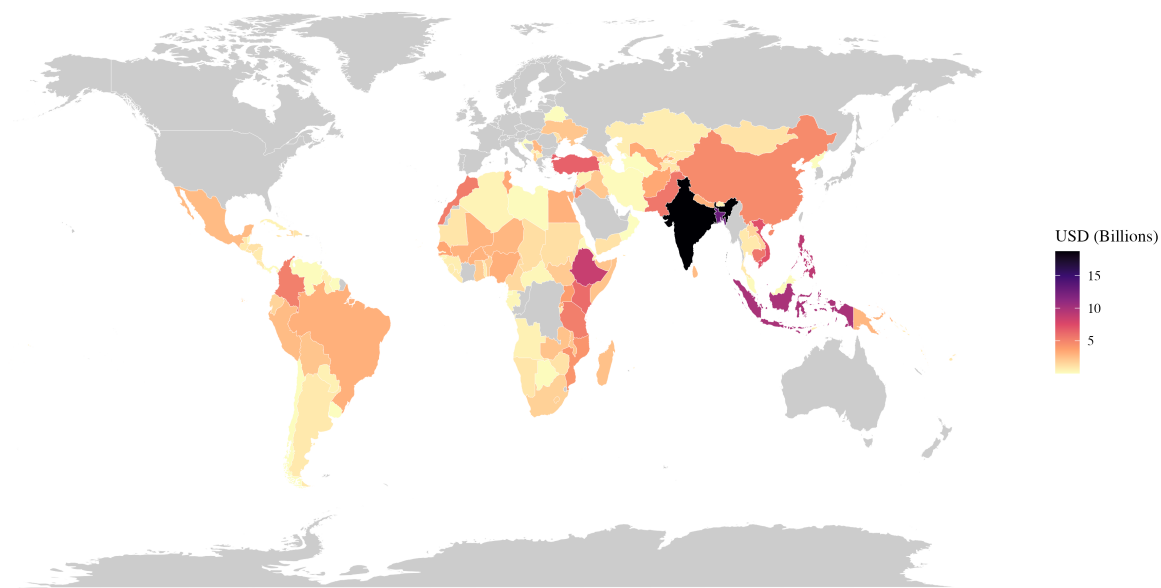


Figure A3: Cumulative adaptation-related commitments by recipient country
Notes: Cumulative adaptation-related commitments received by each country over 2009–2024, in constant USD billion. Darker shading indicates larger cumulative receipts; grey indicates no recorded adaptation-marked flows. Source: OECD Creditor Reporting System, Rio adaptation marker.

A.2.3 NAP adoption and sample coverage

Treatment is defined as the first official submission of a National Adaptation Plan to the UNFCCC. Figure [A4](#) maps NAP submission status at the end of the sample period.

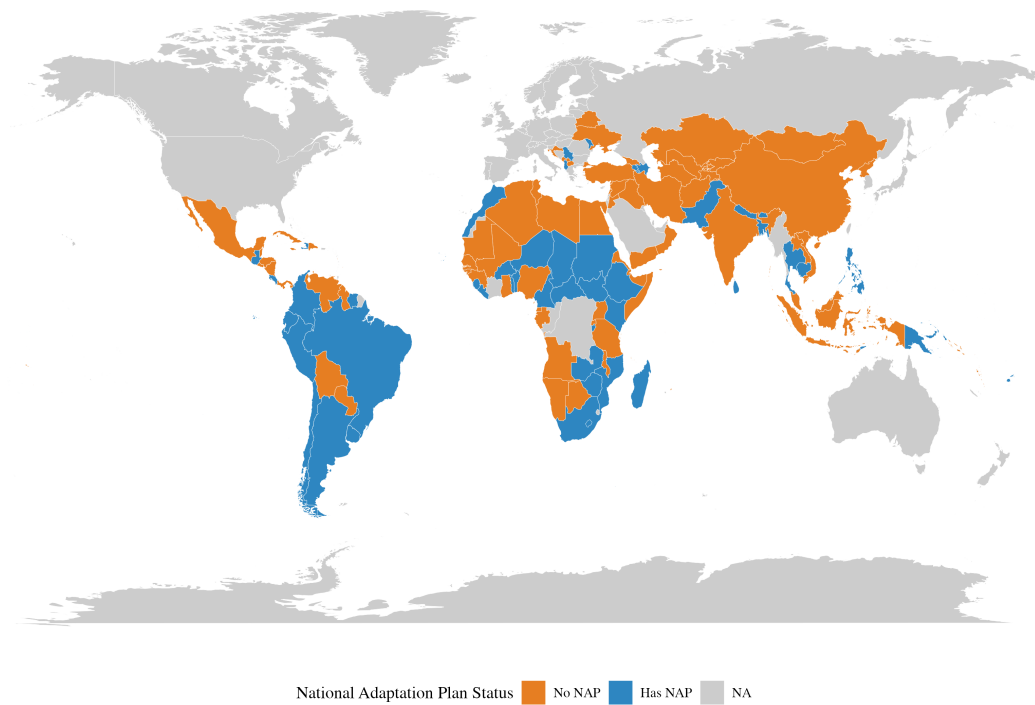


Figure A4: NAP submission status by country

Notes: Countries are classified according to whether they had submitted a National Adaptation Plan to the UNFCCC by the end of 2024. Source: UNFCCC NAP Central tracking tool.

Table A2 reports the distribution of adopters across adoption years and indicates which cohorts enter the main specification, which retains cohorts with at least five treated units.

Table A2: NAP adoption cohorts

Adoption year	Number of adopters	Included in main specification
2015	2	No
2016	4	No
2017	2	No
2018	4	No
2019	4	No
2020	2	No
2021	11	Yes
2022	10	Yes
2023	11	Yes
2024	8	Yes
Total adopters	58	

Notes: The table reports the number of countries first submitting a National Adaptation Plan in each adoption year, among the 58 adopters in the estimation sample; a further 86 never-adopting countries serve as controls. The main specification retains adoption cohorts with at least 5 treated units (2021–2024); the retained-cohorts robustness specification (Section 5.3) additionally includes the smaller 2015–2020 cohorts.

Source: UNFCCC NAP Central tracking tool.

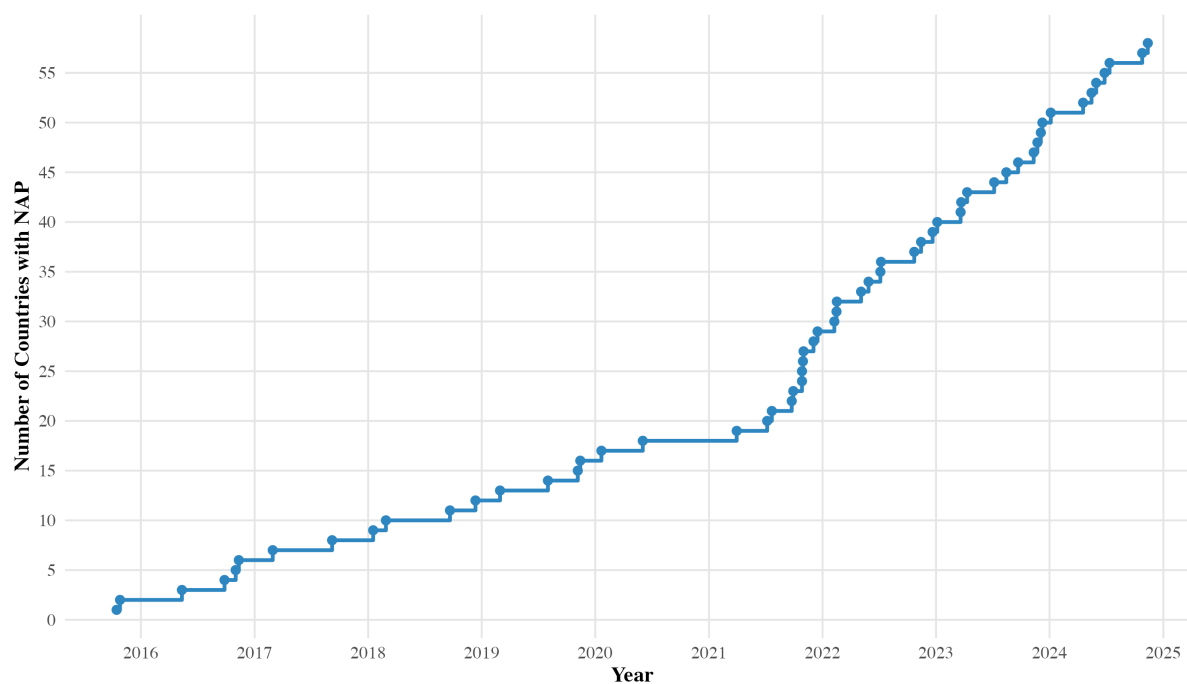


Figure A5: Timing of NAP adoption by country. *Notes:* Each row marks the year in which a country first submitted its National Adaptation Plan to the UNFCCC; countries are ordered by submission date. Source: UNFCCC NAP Central tracking tool.

Table A3: Regional Distribution of NAP Adoption

Region	Countries	First adoption	Last adoption	Median year
Sub-Saharan Africa	21	2015	2024	2022
Latin America & Caribbean	15	2016	2024	2020
Middle East & North Africa	2	2016	2024	2020
South Asia	5	2016	2023	2023
East Asia & Pacific	9	2018	2024	2021
Europe & Central Asia	6	2021	2024	2023

Notes: Number of countries having submitted a National Adaptation Plan to the UNFCCC by World Bank region, with the first, last, and median year of submission in each region.

Source: UNFCCC NAP Central tracking tool.

A.2.4 Changes in adaptation finance after NAP adoption

Table A4: Changes in Adaptation Finance After NAP Adoption

Countries	Pre-NAP mean (USD Bn)	Post-NAP mean (USD Bn)	Change (%)
Fiji	0.02	0.11	505.8
Moldova	0.06	0.30	428.9
Togo	0.02	0.08	319.1
Saint Vincent and the Grenadines	0.01	0.02	293.4
Zimbabwe	0.06	0.23	275.0
Tonga	0.01	0.04	253.4
Bangladesh	0.62	2.17	249.3
Sierra Leone	0.03	0.11	245.7
Democratic Republic of the Congo	0.13	0.44	242.1
Philippines	0.52	1.76	235.9
Costa Rica	0.04	0.12	228.8
Sri Lanka	0.08	0.24	216.8
Madagascar	0.11	0.35	216.2
Zambia	0.12	0.38	213.3
Brazil	0.09	0.28	212.1
Burkina Faso	0.08	0.25	196.6
Sudan	0.04	0.10	176.7
Serbia	0.17	0.43	153.3
South Africa	0.09	0.21	143.9
Central African Republic	0.01	0.04	143.9
Morocco	0.31	0.73	138.1
Colombia	0.21	0.50	136.6
Cambodia	0.28	0.65	134.6
Timor-Leste	0.03	0.07	124.9
Benin	0.10	0.22	115.8
Mozambique	0.25	0.52	113.2
Papua New Guinea	0.16	0.34	107.6
Chad	0.07	0.14	102.0
Cameroon	0.06	0.12	83.8
Ethiopia	0.43	0.79	83.7
Niger	0.16	0.28	77.1
Liberia	0.04	0.07	68.9
Albania	0.10	0.17	68.9
Burundi	0.06	0.09	64.9
Argentina	0.05	0.08	64.3
Pakistan	0.34	0.56	62.5
West Bank and Gaza Strip	0.11	0.17	62.3
Thailand	0.05	0.08	59.2
Kenya	0.31	0.49	54.5
Guatemala	0.05	0.07	38.9
Chile	0.00	0.01	37.7
Armenia	0.06	0.08	33.6
Saint Lucia	0.01	0.01	33.5
Marshall Islands	0.01	0.01	25.8
Peru	0.16	0.20	23.5
Kiribati	0.02	0.03	22.6
South Sudan	0.14	0.17	21.7
Haiti	0.10	0.12	21.5
Nepal	0.19	0.21	5.8
Cabo Verde	0.03	0.02	-23.5

Notes: Mean annual adaptation-related commitments (USD billion, constant prices) received by each NAP-adopting country before and after its adoption year; the final column is the percentage change of the post-adoption mean relative to the pre-adoption mean. Descriptive only — these raw changes do not adjust for time trends or composition.

Sources: OECD CRS (Disaster Preparedness); UNEFCC NAP Catalogue

A.2.5 List of donors

Table A5: List of donors by donor type

Donor Type	Donor Name
Bilateral_members	Australia
Bilateral_members	Austria
Bilateral_members	Azerbaijan
Bilateral_members	Belgium
Bilateral_members	Canada
Bilateral_members	Croatia
Bilateral_members	Czechia
Bilateral_members	Denmark
Bilateral_members	EU Institutions
Bilateral_members	Estonia
Bilateral_members	Finland
Bilateral_members	France
Bilateral_members	Germany
Bilateral_members	Greece
Bilateral_members	Hungary
Bilateral_members	Iceland
Bilateral_members	Ireland
Bilateral_members	Italy
Bilateral_members	Japan
Bilateral_members	Kazakhstan
Bilateral_members	Korea
Bilateral_members	Kuwait
Bilateral_members	Latvia
Bilateral_members	Liechtenstein
Bilateral_members	Lithuania
Bilateral_members	Luxembourg
Bilateral_members	Netherlands
Bilateral_members	New Zealand
Bilateral_members	Norway
Bilateral_members	Poland
Bilateral_members	Portugal
Bilateral_members	Qatar
Bilateral_members	Romania
Bilateral_members	Slovak Republic
Bilateral_members	Slovenia
Bilateral_members	Spain
Bilateral_members	Sweden
Bilateral_members	Switzerland
Bilateral_members	United Arab Emirates
Bilateral_members	United Kingdom

Continued on next page

Table A5 – continued from previous page

Donor Type	Donor Name
Bilateral_members	United States
Multilateral_donors	Adaptation Fund
Multilateral_donors	Arab Fund (AFESD)
Multilateral_donors	Asian Development Bank
Multilateral_donors	Caribbean Development Bank
Multilateral_donors	Central Emergency Response Fund
Multilateral_donors	Climate Investment Funds
Multilateral_donors	Council of Europe Development Bank
Multilateral_donors	European Bank for Reconstruction and Development
Multilateral_donors	Food and Agriculture Organisation
Multilateral_donors	Global Environment Facility
Multilateral_donors	Green Climate Fund
Multilateral_donors	IFAD
Multilateral_donors	IMF Resilience and Sustainability Trust
Multilateral_donors	International Development Association
Multilateral_donors	International Labour Organisation
Multilateral_donors	Nordic Development Fund
Multilateral_donors	OPEC Fund for International Development
Multilateral_donors	Private Infrastructure Development Group
Multilateral_donors	UN Capital Development Fund
Multilateral_donors	UN Economic and Social Commission for Western Asia
Multilateral_donors	UN Peacebuilding Fund
Multilateral_donors	UNDP
Multilateral_donors	UNECE
Multilateral_donors	UNEP
Multilateral_donors	UNFPA
Multilateral_donors	UNICEF
Multilateral_donors	UNRWA
Multilateral_donors	World Health Organisation
Multilateral_donors	World Tourism Organisation
Other	Arcadia Fund
Other	Asian Forest Cooperation Organisation
Other	Bernard van Leer Foundation
Other	Bezos Earth Fund
Other	Bill & Melinda Gates Foundation
Other	Bloomberg Family Foundation
Other	CGIAR
Other	Carnegie Corporation of New York
Other	Charity Projects Ltd (Comic Relief)
Other	Children's Investment Fund Foundation
Other	Citi Foundation
Other	Conrad N. Hilton Foundation
Other	David and Lucile Packard Foundation

Continued on next page

Table A5 – continued from previous page

Donor Type	Donor Name
Other	Dutch Postcode Lottery
Other	Fondation Botnar
Other	Ford Foundation
Other	Gates Foundation
Other	Gatsby Charitable Foundation
Other	German Postcode Lottery
Other	Global Green Growth Institute
Other	Good Ventures Foundation
Other	Gordon and Betty Moore Foundation
Other	H&M Foundation
Other	Howard G. Buffett Foundation
Other	IKEA Foundation
Other	International Centre for Genetic Engineering and Biotechnology
Other	John D. and Catherine T. MacArthur Foundation
Other	Joint Sustainable Development Goals Fund
Other	La Caixa Banking Foundation
Other	Laudes Foundation
Other	Leona M. and Harry B. Helmsley Charitable Trust
Other	MAVA Foundation
Other	Margaret A. Cargill Foundation
Other	Mastercard Foundation
Other	McKnight Foundation
Other	Monaco
Other	Norwegian Postcode Lottery
Other	Oak Foundation
Other	Open Society Foundations
Other	People's Postcode Lottery
Other	Rockefeller Foundation
Other	Swedish Postcode Lottery
Other	UBS Optimus Foundation
Other	United Nations Conference on Trade and Development
Other	Van Leer Foundation
Other	Wellcome Trust
Other	William and Flora Hewlett Foundation
Other	World Diabetes Foundation

Notes: All providers of adaptation-related finance observed in the estimation sample, grouped into the donor-type categories used in the heterogeneity analysis (DAC and non-DAC bilateral providers, multi-lateral institutions, and private/other providers).

Source: OECD CRS.

A.3 Alternative estimator: De Chaisemartin and d’Haultfoeuille (2024)

We re-estimate the headline effect with the heterogeneity-robust dynamic estimator of De Chaisemartin and d’Haultfoeuille (2024), using the same controls as the main specification (WGI Government Effectiveness, log population) and the same thin-dropped sample. This estimator is robust to heterogeneous and dynamic treatment effects and uses both never-treated and not-yet-treated units as controls, providing an independent check on the Callaway–Sant’Anna design. The average total effect is 0.298 ($SE = 0.122$; 95% CI [0.059, 0.537]), almost identical to the CS simple ATT of 0.294 in Table 2. The dynamic effects build over the post-adoption window (Effect +1 = 0.390, rising to +4 = 0.556), and the placebo (pre-treatment) estimates are individually insignificant, consistent with parallel pre-trends. The close agreement of two heterogeneity-robust estimators indicates that the headline result is not an artefact of the CS estimator.

Table A6: Robustness: de Chaisemartin and D’Haultfoeuille (2024) estimator

	Estimate	SE	95% CI	N
Av. total effect	0.298	(0.122)	[0.059, 0.537]	504
Effect +1	0.390	(0.139)	[0.118, 0.662]	440
Effect +2	0.227	(0.158)	[-0.083, 0.537]	317
Effect +3	0.095	(0.178)	[-0.254, 0.445]	201
Effect +4	0.556	(0.266)	[0.034, 1.077]	97
Placebo -1	0.122	(0.158)	[-0.188, 0.432]	440
Placebo -2	0.134	(0.189)	[-0.236, 0.505]	317
Placebo -3	0.377	(0.215)	[-0.044, 0.799]	201

Notes: de Chaisemartin and D’Haultfoeuille (2024) heterogeneity-robust DID estimator, using a current-treatment dummy (1 if the country has adopted a NAP by year t); never-treated and not-yet-treated countries serve as controls. Same controls as the CS main specification (WGI gov. effectiveness, log population). The average total effect is the average of the dynamic effects; the placebos test for pre-trends. 95% confidence intervals are produced by the package.

Source: OECD CRS (Rio adaptation markers); UNFCCC NAP Central.

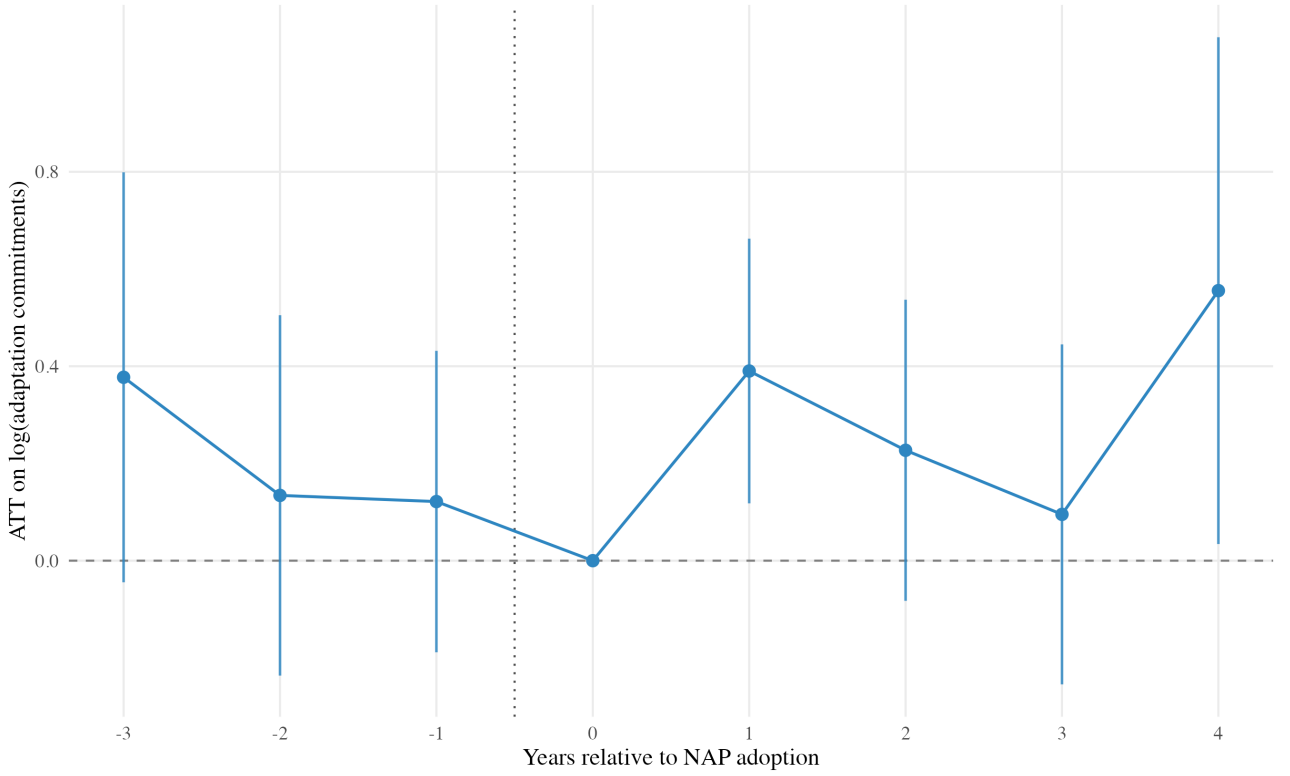


Figure A6: Dynamic ATT of NAP adoption on log adaptation commitments, [De Chaisemartin and d’Haultfoeuille \(2024\)](#) estimator. *Notes:* de Chaisemartin and D’Haultfoeuille heterogeneity-robust dynamic estimator; never-treated and not-yet-treated controls; thin-dropped sample; controls: WGI Government Effectiveness, log population. Placebo (pre-treatment) estimates are individually insignificant, consistent with parallel pre-trends. Source: authors’ estimations on OECD CRS (Rio adaptation markers) and UNFCCC NAP Central data.

A.4 Goodman-Bacon decomposition

To gauge how much the two-way fixed-effects (TWFE) estimate is contaminated by “forbidden” comparisons that use already-treated units as controls, we apply the [Goodman-Bacon \(2021\)](#) decomposition on the balanced-panel subsample. The overwhelming majority of the TWFE weight — 89.6% — comes from clean treated-versus-untreated comparisons (average estimate 0.190), while only 1.2% falls on the problematic “later versus earlier treated” comparisons (Table [A7](#)). Because the forbidden comparisons carry negligible weight, the TWFE estimate (0.175 overall) is not materially biased by negative weighting, which is consistent with the close agreement between the TWFE, CS, and dCDH estimates. This explains why the heterogeneity-robust estimators and the naive TWFE land in the same region, and corroborates the design.

Table A7: Goodman-Bacon decomposition of the two-way fixed-effects estimator

Comparison	Weight	Avg. estimate
Treated vs Untreated	0.8963	0.1896
Earlier vs Later Treated	0.0920	0.0022
Later vs Earlier Treated	0.0116	0.4105
Overall (TWFE)	1.0000	0.1749

Notes: Decomposition of the static two-way fixed-effects estimate of `post_nap` (an indicator for years after NAP adoption) on log adaptation commitments into 2×2 comparison groups (Goodman-Bacon 2021); the weights sum to one. The “Later vs Earlier Treated” group is the problematic forbidden comparison that uses already-treated units as controls. Estimated on the balanced-panel subsample.

Source: OECD CRS (Rio adaptation markers); UNFCCC NAP Central.

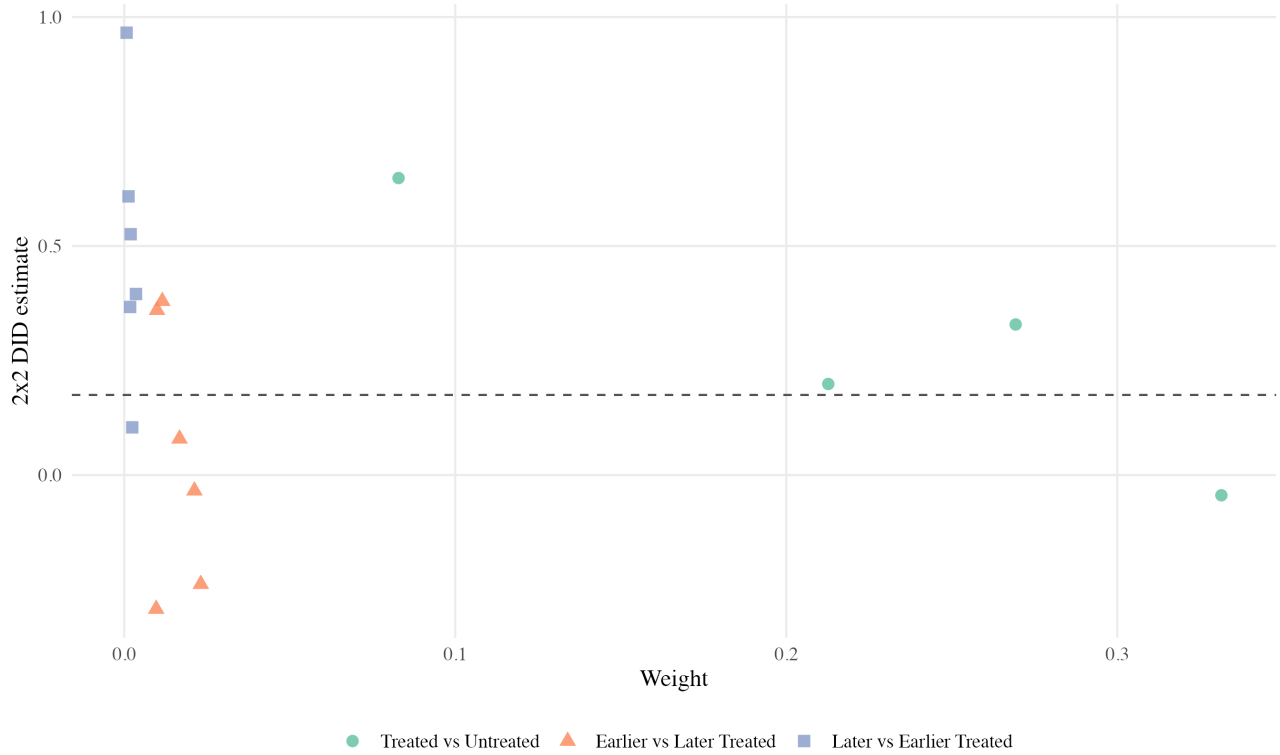


Figure A7: Goodman-Bacon decomposition of the TWFE estimate. *Notes:* Each point is a two-by-two difference-in-differences comparison underlying the TWFE estimate, plotted against its decomposition weight; comparison type distinguishes treated-versus-untreated from later-versus-earlier-treated terms. The forbidden later-versus-earlier comparisons receive negligible weight, so the TWFE estimate is not materially biased by negative weighting. Source: authors’ estimations on OECD CRS (Rio adaptation markers) and UNFCCC NAP Central data.